GINHAWA · AIR-QUALITY STATION · BEGINNER EDITION

Station Build Manual

Everything needed to build one Ginhawa monitoring station from the parts the team bought, written for people who have never built a device. It starts with the basics (what voltage is, how to use a multimeter, how to solder) and then builds the station one tested piece at a time. Do the stages in order and pass every **Check** before moving on.

5 parts **29** stages **31** figures about **5 – 7 days** of work

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PART 1

Learn before you build

Eight short stages with no work on the real parts. Read them together as a team. Everything later in the manual assumes you know what is in here.

STAGE 1 `read first`

What you are building

A solar-powered box that measures the air and reports to the Ginhawa app on its own.

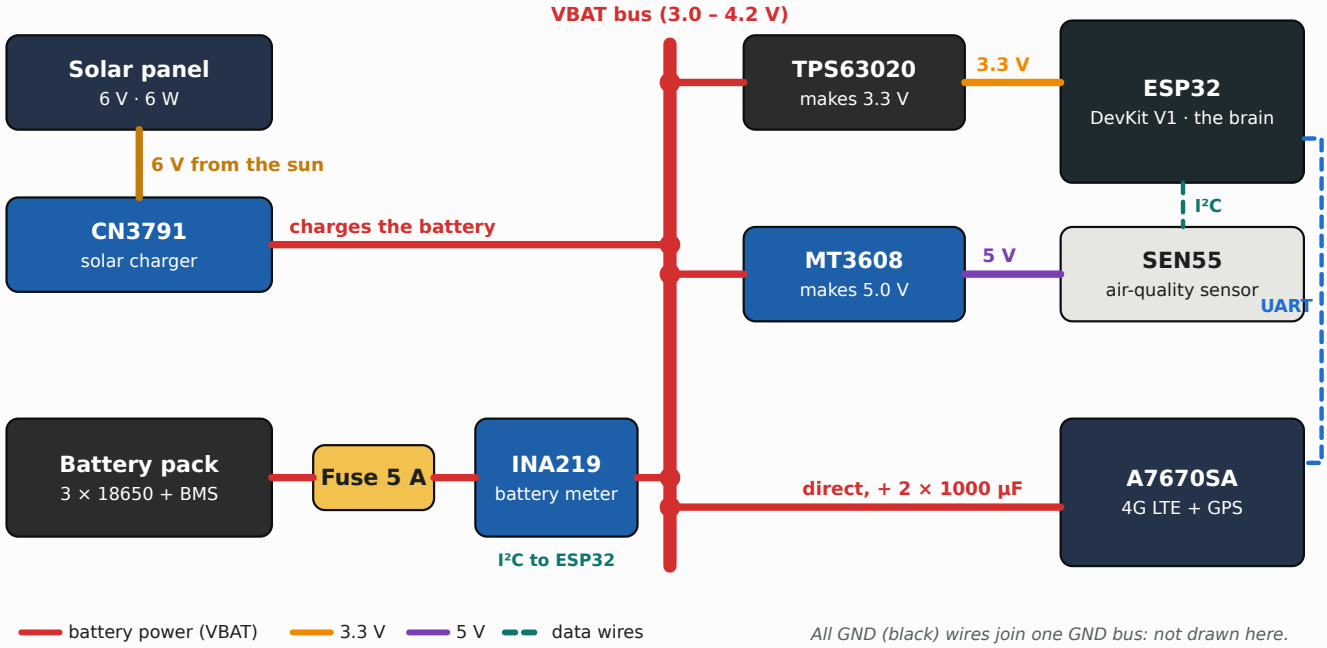


Figure 1. How power and data move through the station. The battery feeds one **VBAT bus**; every module takes its power from that bus. The solar panel refills the battery through the CN3791.

How the station works, in plain words

1. **Sun → battery.** The solar panel makes electricity during the day. The **CN3791** charger uses it to fill three **18650** batteries.
2. **Battery → everything.** The batteries feed one shared power line called the **VBAT bus**. Every part takes its power from there, through a **fuse** that cuts the power if something goes wrong.

3. **Right voltage for each part.** The battery gives 3.0 – 4.2 V depending on how full it is. Two small converters turn that into a steady **3.3 V** (for the ESP32) and **5 V** (for the air sensor).
4. **Measuring.** The **SEN55** measures dust, gases, temperature and humidity every second. The **INA219** measures the battery.
5. **Thinking.** The **ESP32** collects the readings and averages them every minute.
6. **Sending.** The ESP32 sends the readings over **Wi-Fi**. With no Wi-Fi, it uses the **A7670SA 4G modem**. The modem's GPS also tells the app where the station is.

How to use this manual

STOP

Danger to you or to a part. Read it twice.

CAREFUL / IF IT FAILS

Common mistakes, and what to do when a check does not pass.

CHECK

A test with the multimeter. Do not continue until it passes.

TIP

Makes the job easier. Optional.

- **Work in pairs.** One person reads each step out loud and ticks it off; the other does it. Swap every stage.
- **Never skip a Check.** Every Check catches a mistake while it is still cheap to fix.
- **Take a photo after every stage.** It helps when something stops working, and you need photos for the paper anyway.
- **Stuck?** Read "If the check fails" for that stage, then the Troubleshooting table at the end.

Plan

PART	WHAT HAPPENS	TIME
1 Learn	Read, get tools, practise soldering on a spare board	1 day
2 Power system	Battery pack, power board, fuse, charger, converters	1 day
3 Control board	ESP32 board, sensor, modem, test programs	1 – 2 days
4 Software	Start the app, upload the Ginhawa program, 24-hour test	1 – 2 days
5 Box and outdoors	Drill box, install, mount, check in the app	1 day

STAGE 2 45 min

Electricity in plain words

Just enough theory to understand every step. No maths.

WHY THIS MATTERS

When a step says "measure VOUT before connecting" or "+ to + only", this stage is the reason. Understanding it prevents the expensive mistakes.

Voltage, current, power

Think of electricity like water in a pipe:

WORD	UNIT	WATER PICTURE	IN THIS STATION
Voltage	volt (V)	water pressure	Battery 3.0 – 4.2 V. ESP32 3.3 V. Sensor 5 V. Panel 6 V.
Current	ampere (A), milliampere (mA)	how much water flows	Station idle: about 0.1 A. Modem burst: up to 2 A.
Power	watt (W)	pressure × flow	Panel: up to 6 W in full sun.
Capacity	mAh or Wh	size of the tank	Battery pack: 7800 mAh, about 29 Wh: 2 – 3 days without sun.

Plus, minus and ground

Every power source has a **+** (positive) side and a **−** (negative) side. Current flows out of + , through the parts, and back into −. In electronics the − side is called **GND** (ground). **All GND points in the station are connected together**; that is why so many black wires go to one "GND bus".

Polarity means "which way round". The battery, capacitors, the fuse holder's direction (it has none), modules' VIN and VOUT all have a correct way. Connected backwards, most parts are destroyed instantly. That is why this manual uses red for + and black for − everywhere.

Series, parallel, short circuit

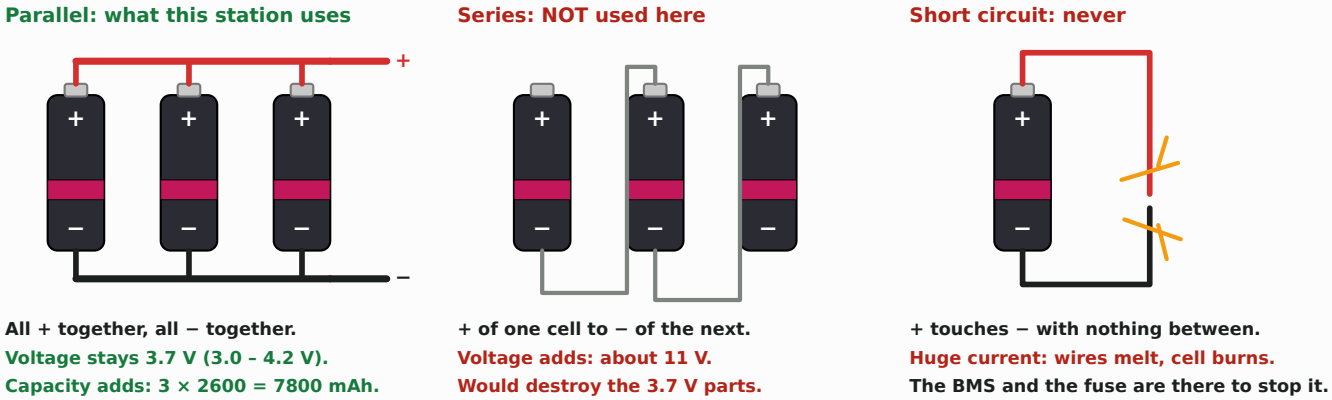


Figure 2. Three ways cells can be connected. The station uses **parallel**: same voltage, more capacity.

Too much voltage destroys parts

Each part has a voltage it accepts. More than that, even for a second, can destroy it. This is why you measure every converter's output **before** connecting anything to it.

PART	ACCEPTS	DESTROYED BY
ESP32 (3V3 pin)	3.0 – 3.6 V	5 V or more
SEN55	4.5 – 5.5 V	more than 5.5 V
A7670SA (VBAT)	3.4 – 4.2 V	more than about 4.4 V
18650 cell	2.5 – 4.2 V	charging above 4.2 V, draining below 2.5 V

Signals: the ESP32 talking to other parts

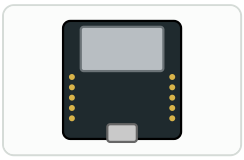
Besides power wires, some wires carry **signals** (information). Two kinds are used:

- **I²C** (say "I-squared-C"): two shared wires, **SDA** (data) and **SCL** (clock). The SEN55 and INA219 both connect to the same two wires. Each has an address so the ESP32 can call them by name: **0x69** is the SEN55, **0x40** the INA219.
- **UART** (serial): two wires, **TX** (transmit) and **RX** (receive). One side's TX goes to the other side's RX. The ESP32 talks to the modem this way.

STAGE 3 30 min

Meet your parts

What each part is, what it does, and how to handle it. Lay all the parts out on a table and find each one.



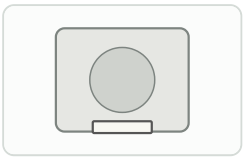
ESP32 DevKit V1

The brain. A tiny computer with Wi-Fi. It runs the Ginhawa program, reads the sensors and sends the readings.

Know: 30 pins in two rows, a USB port for programming, two buttons: EN (restart) and BOOT (used only when uploading).

Careful: Hold it by the edges. Static from your fingers can damage it: touch a metal tap or table leg before handling.

Used in: Stages 17 – 23



Sensirion SEN55

The air-quality sensor: dust (PM1, PM2.5, PM4, PM10), VOC and NOx gases, temperature and humidity. It has a small fan that pulls air through.

Know: One tiny 6-pin connector (JST GH). Two air openings on one side: they must face the outside air.

Careful: Never blow into it, never let it get wet, never block the openings. Most expensive part in the box.

Used in: Stages 20, 26



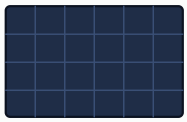
A7670SA LTE + GNSS

The 4G phone modem. It sends readings over the mobile network when there is no Wi-Fi, and its GPS finds the station's location.

Know: Pins for power (VBAT, GND), data (TXD, RXD) and a power key (PWRKEY). Two tiny antenna sockets: MAIN (4G) and GNSS (GPS). A nano-SIM slot.

Careful: The antenna sockets break if pulled sideways. Never power it without the LTE antenna attached.

Used in: Stage 21



Solar panel 6 V 6 W

Turns sunlight into electricity to recharge the battery. Gives about 6 V and up to 1 A in full sun.

Know: Two wires: red (+) and black (−).

Careful: Do not scratch the face. Keep it face down when soldering so it does not make voltage.

Used in: Stages 14, 27



CN3791 solar charger

Takes the panel's 6 V and charges the battery safely, stopping at 4.2 V when full.

Know: Two SOLAR sockets and two BAT sockets (white plugs). Small LEDs show charging / full.

Careful: Must be the 6 V version to match the panel.

Used in: Stage 14



Beston 18650 cell (× 4)

One rechargeable lithium-ion battery, 3.7 V, 2600 mAh. Three of them together power the station for days.

Know: Flat end with the small ring = + (positive). Flat end without = − (negative).

Careful: Never short, puncture, heat or solder it. Store away from metal (keys, coins).

Used in: Stages 10, 11



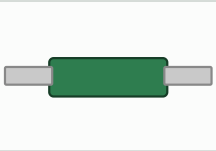
3-cell parallel holder

Holds the three cells and joins them in parallel, so they act like one bigger battery.

Know: Metal plates on one side (+), springs on the other (−). Red and black leads.

Careful: Test it before use: it must be parallel (3.6 – 4.2 V), not series.

Used in: Stage 11



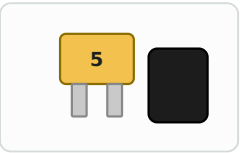
BMS 1S 3.7 V 10 A

Battery protection board. It disconnects the battery if it is over-charged, drained too low, or shorted.

Know: B+ / B− go to the battery. P− (and P+) are the protected output.

Careful: Ground everything to P−, never to B−.

Used in: Stage 11



Fuse 5 A + holder

A thin metal strip that melts and cuts the power if too much current flows. It also works as the station's on/off switch.

Know: The coloured blade fuse plugs into the black waterproof holder.

Careful: Never replace it with a bigger fuse or a wire.

Used in: Stage 13



INA219 power meter

Measures the battery voltage and how much current goes in (charging) or out (using). The app shows this as battery level.

Know: Green screw terminal: VIN+ and VIN− (the battery current flows through these). 6 small pins: VCC, GND, SCL, SDA (plus two not used).

Careful: Tighten the screw terminal well; a loose wire gets hot.

Used in: Stages 13, 20



TPS63020 (3.3 V)

Voltage converter. Turns the battery's changing 3.0 – 4.2 V into a steady 3.3 V for the ESP32.

Know: VIN, GND (input) and VOUT, GND (output).

Careful: Measure its output before connecting anything. It must read 3.3 V.

Used in: Stage 15



MT3608 (5 V)

Voltage booster. Raises the battery voltage to 5 V for the SEN55 sensor.

Know: VIN+, VIN− (input), VOUT+, VOUT− (output), and a blue box with a brass screw to set the voltage.

Careful: Arrives set too high. Set it to 5.0 V before connecting the SEN55.

Used in: Stage 16



1000 µF capacitor (× 20)

A tiny fast-acting energy store. It supplies the modem's short bursts of current so the voltage does not dip.

Know: Long leg = +. The side with the stripe = −.

Careful: Has a direction. Backwards, it can bulge or pop.

Used in: Stage 21



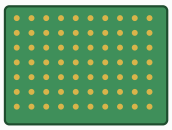
4.7 kΩ resistor (× 50)

Limits current. Two of them "pull up" the sensor data lines so the signals are clean.

Know: Two legs, no direction. Colour bands yellow, violet, red, gold.

Careful: Easy to lose: keep them in their strip until needed.

Used in: Stage 17



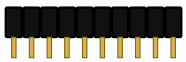
Perfboard (× 3)

A board full of holes with copper rings. Parts and wires are soldered to it instead of a custom circuit board.

Know: Holes 2.54 mm apart. Holes are not connected to each other until you solder a wire between them.

Careful: Use one for practice first.

Used in: Stages 7, 12, 17



Female headers

Sockets soldered to the board so the ESP32 can plug in and be removed.

Know: Strips of 40 sockets, cut to length.

Careful: Cut by pulling a pin first (Stage 8).

Used in: Stage 17



JST GH 6-pin cable

The cable that plugs into the SEN55.

Know: Plug on both ends: you cut it in half and use one half.

Careful: Very small and delicate. Go by pin number, not wire colour.

`Used in: Stage 20`



Silicone wire 18 and 22 AWG

Wire for connections. AWG is the thickness: a smaller number is thicker. 18 AWG for battery power, 22 AWG for everything else.

Know: Red = + (positive). Black = − / GND.

Careful: Silicone does not melt easily, which makes soldering easier.

`Used in: everywhere`



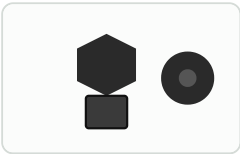
Heat-shrink tubing

Plastic tube that shrinks when heated. Covers solder joints so they cannot touch anything.

Know: Many sizes and colours.

Careful: Slide it on before soldering.

`Used in: everywhere`



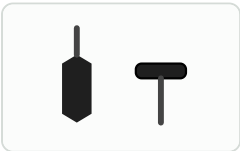
PG7 glands + M12 vent

Waterproof holes for the box. The gland lets the solar cable in; the vent lets air pressure out but keeps water out.

Know: Body, rubber seal and nut.

Careful: Fit on the bottom of the box only.

`Used in: Stage 25`



Standoffs, nuts, screws

Hold the ESP32 board 10 mm off the box wall so nothing under it touches.

Know: M3 standoffs (10 mm), M3 nuts, M3 × 8 screws (2 mm Allen key).

Careful: Nylon does not conduct: safe near circuits.

`Used in: Stage 26`



Silica gel + insect mesh

Silica gel soaks up moisture in the closed box. The mesh keeps insects out of the sensor window.

Careful: Replace silica gel when its beads change colour.

`Used in: Stages 25, 26`

LABEL YOUR BAGS

Write the part name on each bag with a marker as you identify it. Several modules look alike (TPS63020, MT3608, INA219, CN3791).

STAGE 4 1 h

Tools and workspace

What tools you need, what each one is for, and how to set up your table.

WHY THIS MATTERS

Bad tools make every joint harder. A multimeter without a beeper or an iron without a stand turns a 2-day build into a week.

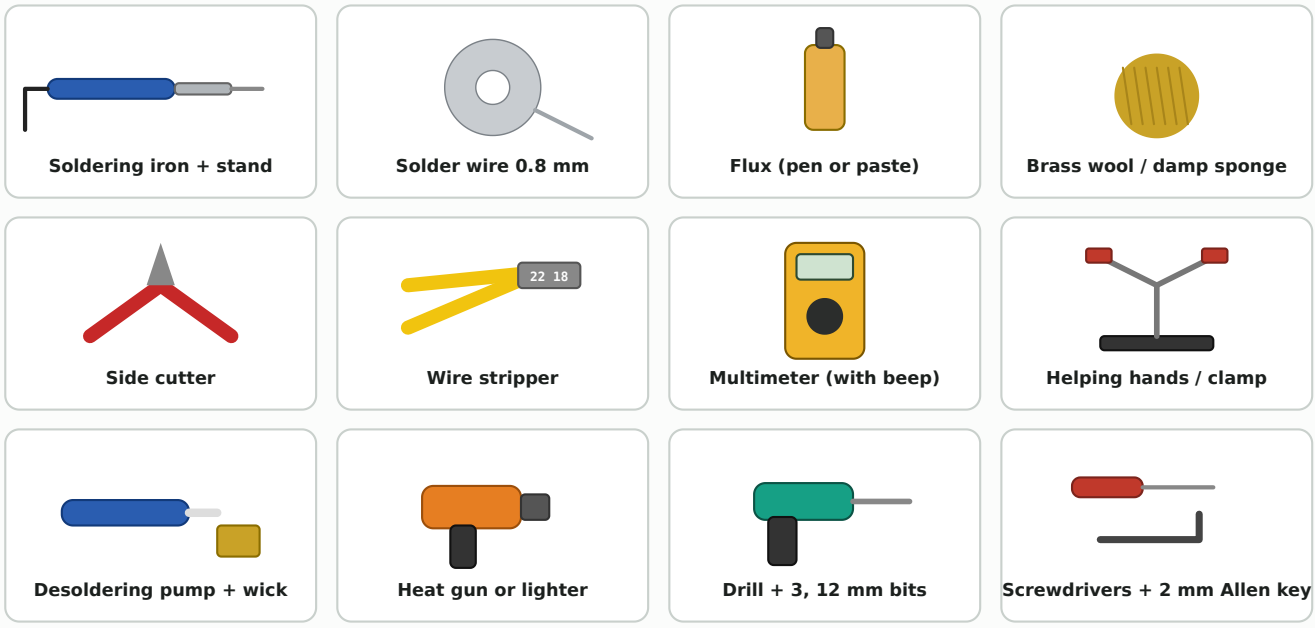


Figure 3. The tools you need. Borrow what you can; the table below says which ones you must have.

TOOL	WHAT IT IS FOR	MUST HAVE?	ABOUT
Soldering iron, adjustable 40 – 60 W, with stand	Melts solder. Adjustable temperature makes it much easier.	Yes	₱300 – 700
Solder wire, 0.8 mm, with flux core (60/40 or 63/37)	The metal that joins things. 0.8 mm is right for small parts.	Yes	₱100 – 250
Flux pen or rosin paste	Makes solder flow onto pads and wires.	Strongly recommended	₱80 – 200
Brass wool tip cleaner or damp sponge	Cleans the iron tip between joints.	Yes	₱80 – 150
Desoldering pump and/or wick (2 mm)	Removes solder to fix mistakes.	Yes	₱100 – 200
Side cutter (flush cutter)	Cuts wire and component legs.	Yes	₱80 – 200
Wire stripper (with 18 and 22 AWG holes)	Removes insulation without cutting the wire strands.	Yes	₱100 – 300
Multimeter **with continuity beep**	Measures voltage, checks connections. Avoid the cheapest DT830B: it has no beeper.	Yes	₱250 – 600
Helping hands or small vise	Holds the board or wires while you solder.	Recommended	₱150 – 300
Heat gun (or lighter)	Shrinks heat-shrink tubing. A lighter works if kept moving.	Yes (either)	₱400+ / ₱20
Drill with 3 mm and 12 mm bits (or step bit)	Holes in the box.	Yes, can borrow	borrow
Small screwdrivers, 2 mm Allen key, pliers	Screw terminals, MT3608 screw, box screws, pulling header pins.	Yes	₱100 – 250
Kitchen scale (grams)	Weighing cells to spot fakes.	Recommended	borrow
Safety glasses	Leg clippings fly when cut.	Yes	₱50 – 100
Masking tape, marker, zip ties, electrical tape, foam tape	Labelling, holding, mounting.	Yes	₱150

Set up your workspace

1. A table with good light, by an open window or with a fan blowing solder smoke away from your face.

2. Cover the table with cardboard or a ceramic tile under the soldering area. Never solder on a bed, carpet or plastic tablecloth.
3. Iron stand on your writing-hand side, cable behind it so you cannot catch it with your arm.
4. A small bowl for cut legs and scraps, and a box for small parts.
5. Keep food and drinks off the table. Wash hands after soldering (solder contains lead).
6. Keep the cells in their own box away from the workspace until a stage tells you to use them.

STAGE 5 10 min

Safety rules

Read aloud as a team before the first soldering session.

LITHIUM BATTERIES

- Never let a bare red and black battery wire touch each other. Tape loose battery wire ends until they are soldered.
- Never solder onto a cell, puncture it, drop it, or leave it in the sun or a hot car.
- Insert the cells **last**, only when a stage tells you to, and take them out before soldering on the power path.
- Store spare cells in a box, never loose in a bag with keys or coins.
- A cell that gets hot, swells, hisses or smells sweet: take it out with pliers, put it outside on concrete away from anything that burns, and do not use it again.

SOLDERING IRON

- The tip is about 350 °C. It burns skin instantly and melts plastic. Always put it in the stand, never on the table.
- Never touch the metal part to see if it is hot. Never catch a falling iron: let it fall.
- Unplug the iron whenever you leave the table, even for 5 minutes.
- If you burn yourself: cool water for 10 minutes. No toothpaste or ice.

FUMES, CLIPPINGS, DRILLING

- Solder smoke irritates eyes and lungs: ventilate, keep your face out of the smoke.
- Wear safety glasses when cutting legs or drilling.
- Wash hands before eating.
- Drill with the box clamped or held firmly on a table, never in your hand.

THE STATION ITSELF IS SAFE TO TOUCH

All voltages in the station are below 7.5 V. They cannot shock you. The dangers are heat (iron, short circuits) and the batteries.

STAGE 6 30 min

Skill: using the multimeter

You will use it at every Check. Practise all three settings now on safe things.

PARTS FOR THIS STAGE

- 1 Beston cell
- 1 AA battery
- 1 resistor
- scrap wire, spare perfboard

TOOLS

- Multimeter

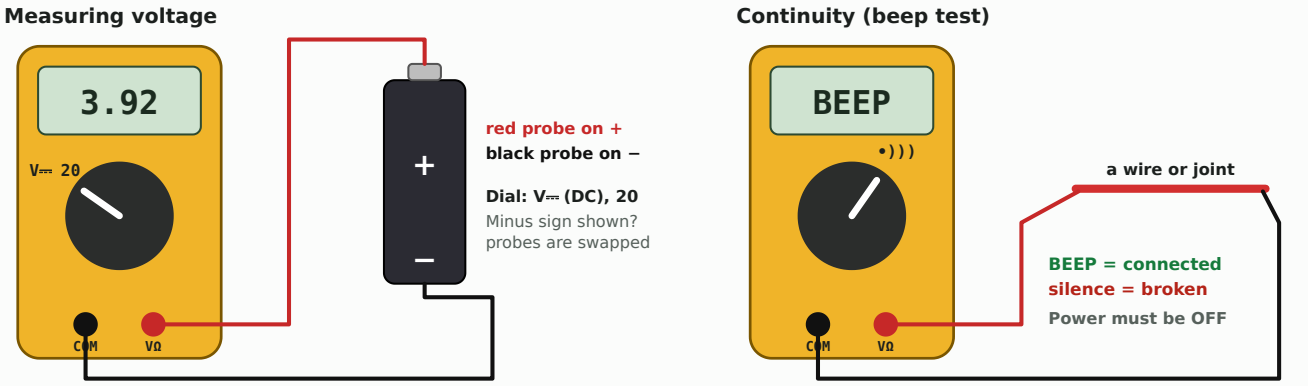


Figure 4. The two multimeter settings you need. Black probe always goes in **COM**, red in **VΩ**.

Before every measurement

1. Black probe in the socket marked **COM**. Red probe in the socket marked **VΩmA** (or V/Ω). Never the 10A socket.
2. Turn the dial to the setting first, then touch the probes.
3. Hold the probes by the plastic. Touch metal tips firmly on the metal you measure.

Setting 1: DC voltage (V⎓ or DCV, range 20)

For measuring a battery, the buses, and converter outputs. Power is **on**.

1 Turn the dial to **20** in the **V⎓** section (straight line with dashes, not the wavy "~" one). On an auto-ranging meter just choose V⎓.

2 Practice: measure a normal AA battery. Red on the bump (+), black on the flat end (−). It reads about 1.5 (new) or 1.2 – 1.4 (used).

3 Swap the probes. It reads −1.5. A minus sign always means the probes are the wrong way round. Nothing is damaged.

4 Practice: measure one Beston cell. It reads between 3.0 and 4.2.

Setting 2: continuity (beep)

For checking that a wire or joint connects, and that two things do NOT connect. Power is **off** (cells out, fuse out).

1 Turn the dial to the symbol that looks like a sound wave or diode (•))) or ▶|). Touch the two probe tips together: it must beep. If not, your meter has no beeper.

2 Practice: touch both ends of a piece of wire. Beep.

3 Practice: touch two separate holes on an empty perfboard. No beep: the holes are not connected.

Setting 3: resistance (Ω, range 20k)

1 Turn the dial to **20k** in the Ω section.

2 Touch the probes to the two legs of one 4.7 kΩ resistor (fingers off the legs). It reads 4.6 – 4.8 (that is 4.6 – 4.8 kΩ).

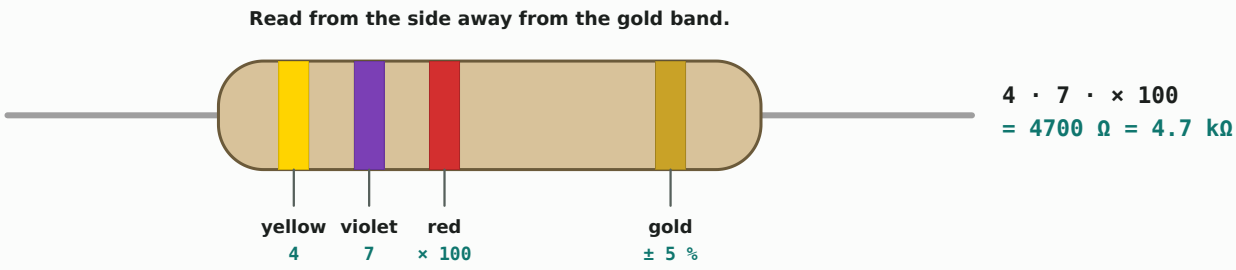


Figure 5. How to read the 4.7 kΩ resistor: **yellow, violet, red, gold**. Or measure it: the multimeter on 20k reads about 4.6 – 4.8.

STAGE 7 1 – 2 h

Skill: soldering

Learn and practise on a spare board before you touch the real parts.

WHY THIS MATTERS

Almost every fault in a home-made device is a bad solder joint. One hour of practice saves days of fault-finding.

PARTS FOR THIS STAGE

- Spare perfboard
- 22 AWG wire
- 2 – 3 resistors
- a few header pins

TOOLS

- Iron, solder, flux, cutter, stripper, multimeter

Set up the iron

1. Put the iron in its stand, plug it in, set it to **350 °C** (60/40 or 63/37 solder) or **370 °C** (lead-free). No temperature control: wait 3 minutes.
2. Wet the sponge (damp, not dripping) or use the brass wool.
3. Melt a little solder onto the tip, then wipe it. The tip should be shiny silver. A black, dull tip will not heat the joint: clean and re-tin it.
4. Re-tin the tip every few joints and before putting it back in the stand.

The golden rule

Heat the parts, not the solder. Touch the tip to the pad and the leg or wire together for 1 – 2 seconds, then feed solder into the joint (not onto the tip). The hot parts melt the solder and it flows over them. Remove solder, then iron. Do not move anything for 3 seconds.

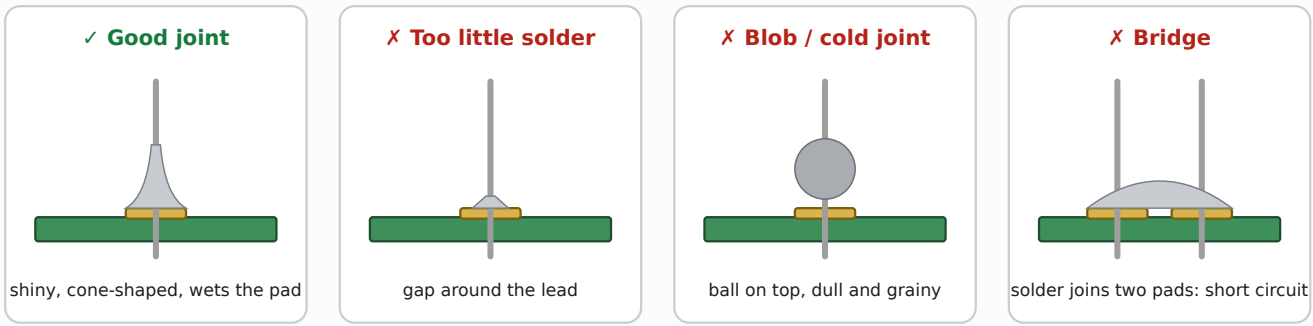


Figure 6. What solder joints should look like. Fix any ✗ joint: reheat it and add or remove solder (use the tip to drag extra away).

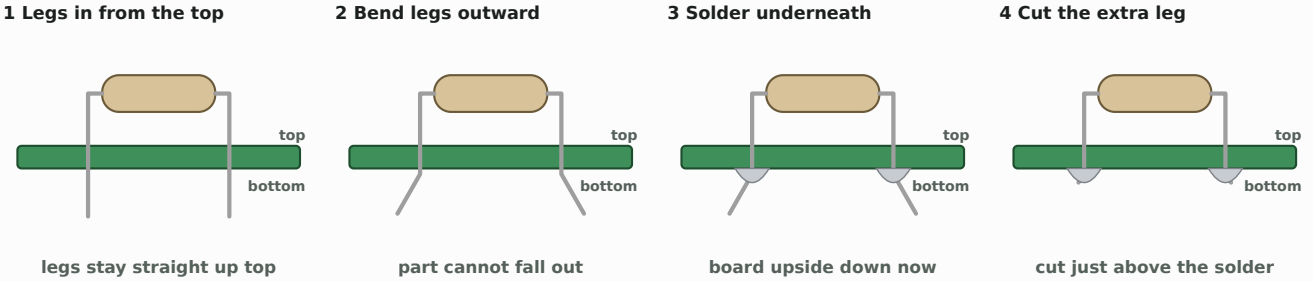


Figure 7. Soldering a part with legs (resistors, capacitors, headers) through the perfboard holes.

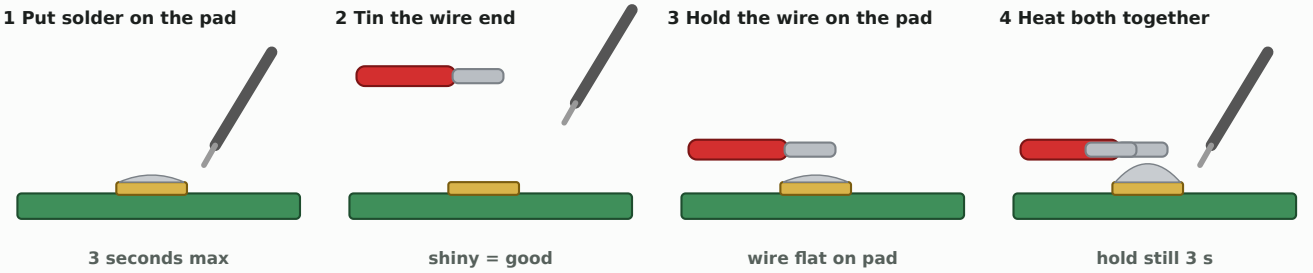


Figure 8. Soldering a wire onto a flat pad (module pads, BMS strips, bus wires). Tinning both parts first makes it quick and strong.

Fixing mistakes

- **Too much solder or a bridge:** clean the tip, touch it to the blob, and drag the excess away; or use the desoldering pump (press down, put the nozzle on the melted joint, press the button) or wick (lay it on the joint, press with the iron, the copper soaks up the solder).
- **Dull, grainy joint (cold joint):** add a little flux and reheat until it flows and turns shiny.

- **Pad lifted off the board:** stop heating that spot; use the next hole and run a short wire.

Practice board

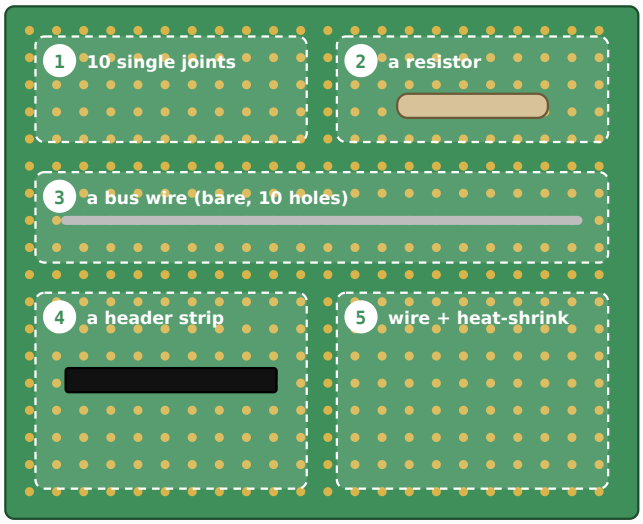


Figure 9. Practice board: five exercises before you touch the real parts.

1. Exercise 1: solder 10 short pieces of wire leg (cut resistor legs) into 10 holes. Compare each joint with Figure 6.

2. Exercise 2: solder one resistor through two holes, legs bent outward, then cut the legs (Figure 7).

3. Exercise 3: make a bus wire: strip 7 cm of 22 AWG completely, tin it, lay it along a row and solder every 4th hole.

4. Exercise 4: solder a 6-pin piece of female header straight. Solder one end pin, check it is straight, then the others.

5. Exercise 5: strip, tin and solder a wire end onto the bus wire, with heat-shrink (Stage 8 shows how).

6. Check every joint with the continuity setting: wire end to bus beeps; neighbouring separate holes do not beep.

IF THE CHECK FAILS

- Solder balls up and does not stick: the pad is not hot enough or dirty. Add flux, hold the iron 1 second longer.
- Solder smokes and burns black: iron too hot or held too long. Lower the temperature, work faster.
- Iron does not melt solder at all: tip is oxidised. Clean on brass wool, re-tin, or replace the tip.

STAGE 8 45 min

Skill: wires, heat-shrink, headers and connectors

Preparing wires correctly is half of a reliable build.

PARTS FOR THIS STAGE

- scrap 18 and 22 AWG wire
- heat-shrink
- a female header strip

TOOLS

- Stripper, cutter, pliers, iron, heat gun or lighter



Wire stripper holes: put the wire in the hole with the SAME number as the wire (AWG).
18 AWG (power wire) → hole marked 18 or 1.0 mm. 22 AWG (signal wire) → hole marked 22 or 0.6 mm.
Hole too small cuts strands (wire gets weak). Hole too big only scrapes the insulation. Count the strands: none should be cut.

Figure 10. Preparing every wire end. A tinned end looks silver and stiff and never frays.

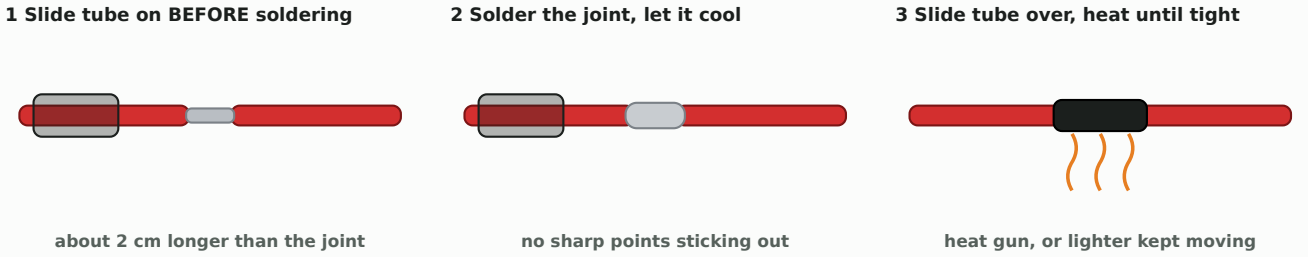


Figure 11. Heat-shrink tubing insulates a joint. It shrinks to about half its size when heated.

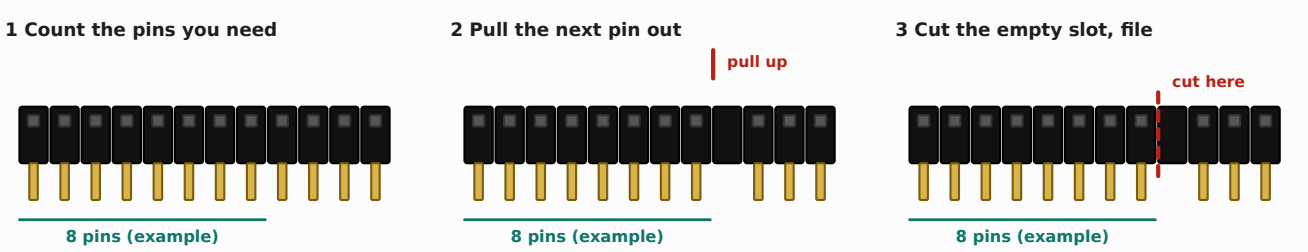


Figure 12. Cutting female headers to length. You lose one pin at each cut, so pull that pin out first: the plastic then cuts cleanly.

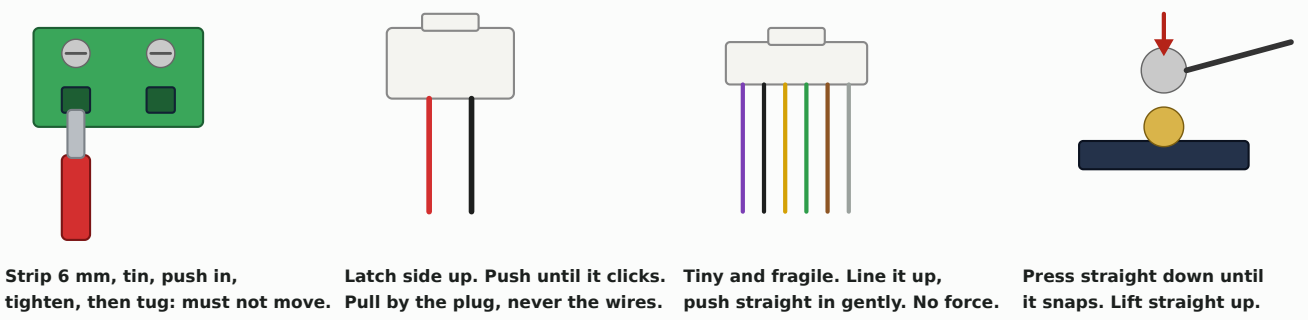


Figure 13. The four kinds of connectors in the station and how to handle each one.

Wire lengths and colours

- Cut every wire about **2 cm longer** than the distance. Too long is easy to fix; too short means starting again.
- **Red** for every + and voltage wire, **black** for every GND. You only have red and black, so mark signal wires at both ends with a coloured heat-shrink piece: **yellow = SDA**, **green = SCL**, **blue = modem TX/RX**, **orange tag = 3.3 V**, **purple/other tag = 5 V**.
- Twist pairs that go together (for example SDA and SCL) loosely: tidier and less noise.

PART 2

Build the power system

Battery, protection, charger and the two voltage converters. At the end, you have a working power supply with no computer attached yet.

STAGE 9 45 min

Check every part when it arrives

Most sellers give 7 days for returns. Test each part the day it arrives, before you solder anything to it.

PARTS FOR THIS STAGE

- every part as it arrives

TOOLS

- Multimeter
- kitchen scale

PART	HOW TO CHECK	RETURN IT IF
A7670SA breakout	Read the metal shield on the module: it must say **A7670SA-FASE** (FASE = with GPS). Two antenna sockets. Two antennas in the bag.	Says LASE or LASC
Beston 18650 × 4	Weigh each (kitchen scale): about **46 g**. Measure each (V⎓ 20): between 3.0 and 4.2 V.	Under 40 g, or below 2.5 V
3-cell holder	Look: all springs on the same side. Later with cells in: leads read 3.6 – 4.2 V.	Springs alternate, or reads 11 – 12 V
Solar panel	In direct sun, measure the red and black wires (V⎓ 20): **6 – 7.5 V**. Cover it with your hand: drops.	Below 5 V in full sun
CN3791	The listing option was **6V**. A sticker or marking may show 6V. Comes with plug cables.	Wrong version
TPS63020	Option was **3.3V**. Tested in Stage 15.	Output not 3.3 V
MT3608	Blue board with a blue box and brass screw.	Damaged
BMS	Printed **3.7V** or the listing said 3.7V-10A. Pads B+, B−, P− (and maybe P+).	Says 3.2V / LFP
INA219	Blue board, big resistor marked **R100**, green screw terminal and pin strip in the bag.	Missing parts
JST GH cable	6 wires, very small plug. Hold it next to the SEN55 socket: same size.	Bigger plug (XH or PH)
Resistors	Measure one: 4.6 – 4.8 kΩ.	Different value
Capacitors	Printed 1000 µF, 25 V. Stripe on the side.	Different value
Fuses	Standard size blade, 5 A (and 7.5 A) present. Holder is standard size, fuse fits.	Size mismatch

KEEP RECEIPTS AND PHOTOS

Screenshot each order. If a part fails a check, message the seller the same day with a photo of the reading.

STAGE 10 15 min

Prepare the battery cells

Cells joined in parallel must start at almost the same voltage.

WHY THIS MATTERS

If a full cell is connected to an empty one, the full one pushes a very large current into the empty one the moment they touch. That damages both and can make them hot.

PARTS FOR THIS STAGE

- 4 Beston cells
- masking tape, marker

TOOLS

- Multimeter

What the voltage tells you

CELL VOLTAGE	MEANING
4.15 – 4.20 V	Full
3.8 – 3.9 V	About 60 %
3.6 – 3.7 V	About 30 – 40 %: normal for new cells from the shop
3.3 – 3.4 V	Nearly empty
below 3.0 V	Empty: charge soon
below 2.5 V	Damaged: do not use

1. Measure each of the 4 cells. Write its voltage on a piece of masking tape and stick it on the cell.

2. Pick the **3 cells** whose numbers are closest together. Mark the 4th "SPARE" and store it in a box.

3. If all 3 are within **0.05 V** of each other (example: 3.71, 3.72, 3.74), they are ready. Go to the next stage.

4. If not: continue the build without cells. In Stage 14 (charger) you will charge each cell alone until it reads 4.15 – 4.20 V. Then all 3 are equal.

STAGE 11 45 min

Build the battery pack

Connect the holder to the BMS protection board.

PARTS FOR THIS STAGE

- Holder
- BMS
- 3 matched cells
- 18 AWG red and black
- heat-shrink

TOOLS

- Iron
- cutter
- stripper
- heat gun
- multimeter

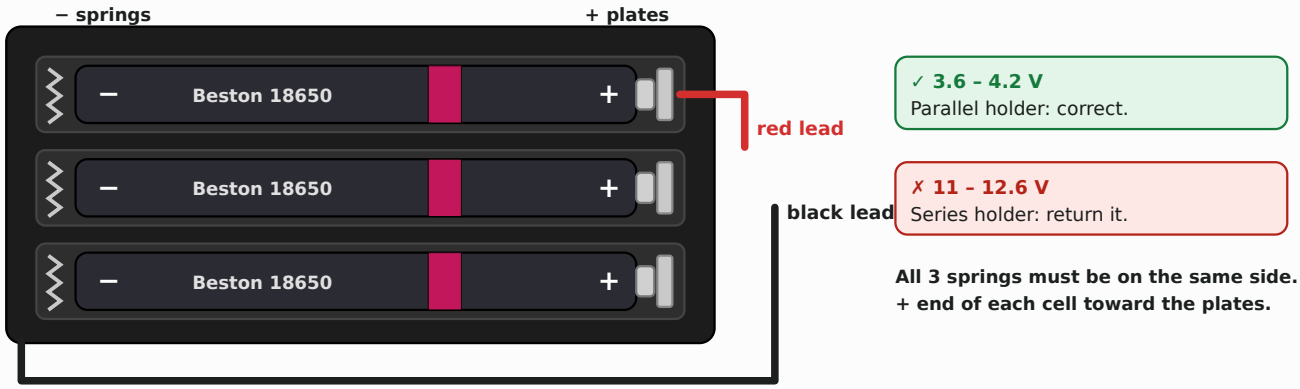


Figure 14. Cells in the 3-cell **parallel** holder. Every cell faces the same way: + end on the metal plates, − end on the springs.

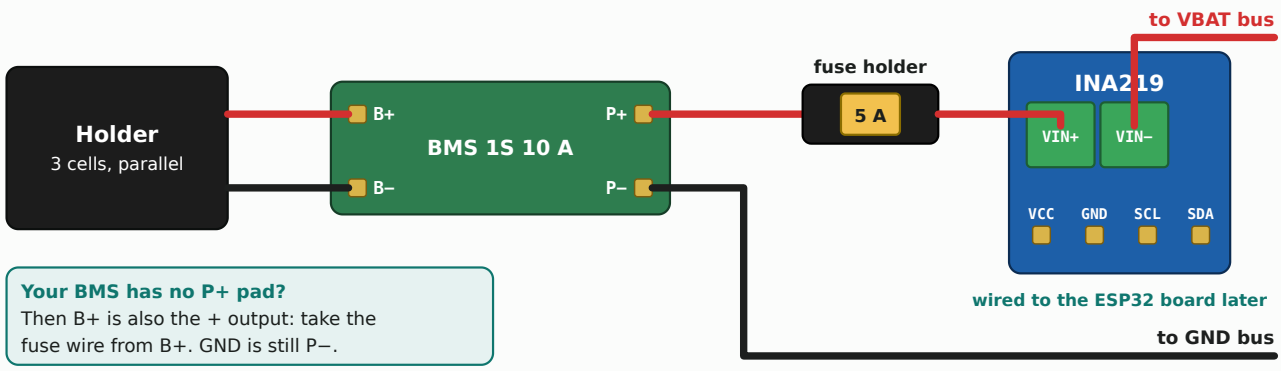


Figure 15. Battery → BMS → fuse → INA219. The thick red path carries all the station's current. **P−** becomes the station's ground (GND).

1 Make sure there are **no cells** in the holder.

2 Cut the holder's thin red and black leads to 5 cm. Strip 5 mm and tin both.

3 Cut two 10 cm pieces of 18 AWG wire, one red, one black. Strip and tin both ends of each. Put a 3 cm piece of heat-shrink on each.

4 Join holder red lead to the 18 AWG red wire: hold the tinned ends side by side, heat until the solder flows together. Slide the heat-shrink over and shrink it. Same for black.

5 Look at the BMS. Find the printed pads **B+**, **B−**, **P−** and maybe **P+** (use a phone camera zoom if they are small). Photograph it.

6 Cut the BMS nickel strips to about 5 mm with the side cutter. Tin them.

7 Solder the red wire from the holder to **B+**. Solder the black wire to **B−** (wire-to-pad method, Figure 8).

8 Cut a 15 cm red 18 AWG wire. Solder it to **P+**, or to B+ next to the first wire if your BMS has no P+.

9 Cut a 15 cm black 18 AWG wire. Solder it to **P−**. This is the station's main ground.

10 Wrap the free ends of these two new wires separately in electrical tape.

11 Wrap the BMS in a piece of heat-shrink or electrical tape so no pad can touch metal.

12 Insert the 3 matched cells: flat + end against the metal plates, − end against the springs.

CHECK

MEASURE BETWEEN	MULTIMETER ON	EXPECTED
Holder red and black leads (at the BMS B+ and B−)	V⎓ 20	**3.6 – 4.2 V**
P+ (or B+) and P− wire ends	V⎓ 20	**same as above**

GROUNDS GO TO P−, NEVER B−

A ground wire connected to B− bypasses the BMS: the battery would no longer be protected.

NOW REMOVE THE CELLS

Take the cells out until Stage 13. The next stages solder on the power path.

IF THE CHECK FAILS

- P+ to P− reads 0 V but the cells read fine: the BMS is "asleep". It wakes when the charger is connected (Stage 14). Continue.
- Reads about 11 V: the holder is wired in series. Stop, take the cells out, return the holder.
- Nothing at all on the holder leads: a cell is in backwards or a lead joint is broken. Check each cell direction and the joints with continuity.

STAGE 12 30 min

Make the power board

A small perfboard with two long bare wires: one for battery power (VBAT bus), one for ground (GND bus).

WHY THIS MATTERS

Instead of joining many wires in mid-air, every part simply solders to a bus. It is neater, stronger, and each connection can be checked.

PARTS FOR THIS STAGE

- 5 × 7 cm perfboard
- 18 AWG wire (14 cm)

TOOLS

- Iron
- stripper
- pliers
- marker
- multimeter

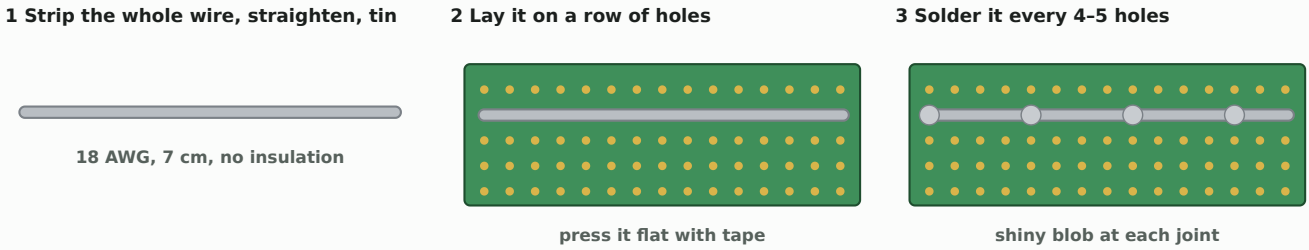


Figure 16. Making a bus wire: one long bare wire that many other wires solder onto.

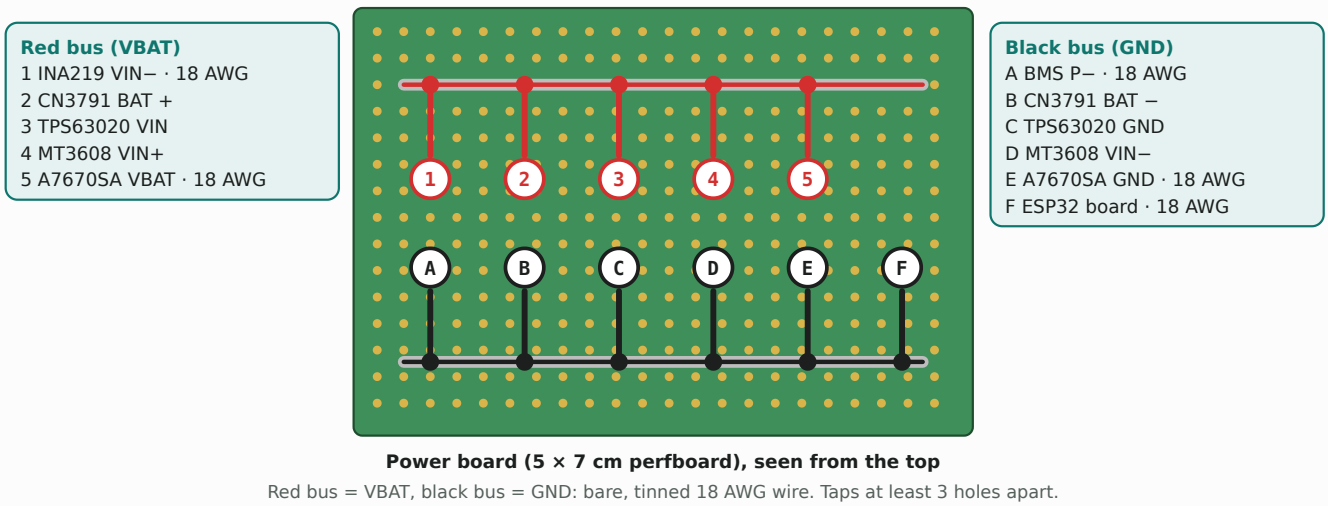


Figure 17. The power board is two long bus wires. Everything that needs battery power solders to the red bus (taps 1–5); every ground solders to the black bus (taps A–F).

1. Take the **5 × 7 cm** perfboard. Decide which side is the top (the side without copper rings, if only one side has them).

2. Cut two 7 cm pieces of 18 AWG wire. Strip the **whole** insulation off both. Straighten them by pulling them gently with two pliers.

3. Tin both bare wires completely.

4. Lay the first along the second row of holes from the top. Hold it with tape. Solder it to every 4th hole. This is the **VBAT bus**: mark it with red marker on the board.

5. Lay the second along the second row from the bottom and solder it the same way. This is the **GND bus**: mark it black.

6. Write the tap numbers 1 – 5 and letters A – F on the board with a marker, as in Figure 17. Later each wire goes to its numbered spot.

CHECK (NO POWER CONNECTED)

MEASURE BETWEEN	MULTIMETER ON	EXPECTED
VBAT bus and GND bus	continuity	**NO beep**
Left end and right end of the VBAT bus	continuity	**beep**
Left end and right end of the GND bus	continuity	**beep**

IF THE CHECK FAILS

- VBAT and GND beep together: solder has bridged the two rows somewhere, or a strand touches. Look along both buses with a light; remove the bridge with the iron or wick.

STAGE 13 45 min

Fuse and battery meter (INA219)

The battery's + wire goes through the fuse and the INA219 before reaching the VBAT bus.

PARTS FOR THIS STAGE

- INA219 (with its terminal and pins)
- fuse holder
- 5 A fuse
- 18 AWG red

TOOLS

- Iron
- small screwdriver
- multimeter

Follow the right half of Figure 15.

1. Solder the INA219's green screw terminal onto the INA219 board in the two big holes marked VIN+ and VIN− (openings facing outward). Solder the 6-pin header strip in the row of 6 small holes, short pins through the board.

2. Take the fuse holder. Make sure it is empty (no fuse in it). Its two wires have no direction.

3. Strip and tin both fuse holder wire ends. Slide heat-shrink onto the BMS **P+** wire. Join P+ to one fuse holder wire (side by side, solder, shrink).

4. Push the other fuse holder wire end into the **VIN+** opening of the INA219 terminal. Tighten the screw firmly. Tug the wire: it must not move.

5. Cut 10 cm of red 18 AWG. Strip both ends. Screw one end into **VIN−**. Solder the other end to the VBAT bus at **tap 1**.

6. Solder the BMS **P−** black wire to the GND bus at **tap A**.

7. Keep the fuse out of the holder. From now on: **fuse out = everything off**.

CHECK: PUT THE CELLS IN, THEN PUSH THE 5 A FUSE INTO THE HOLDER

MEASURE BETWEEN	MULTIMETER ON	EXPECTED
VBAT bus and GND bus	V⎓ 20	**3.6 – 4.2 V (same as the battery)**

Then pull the fuse out again.

IF THE CHECK FAILS

- 0 V on the buses: the BMS may still be asleep (it wakes in Stage 14), the fuse is not pushed in fully, or a screw terminal is loose. Measure step by step: P+ to P−, then fuse holder out side to P−, then VIN− to P−.
- The fuse blew immediately: there is a short. Fuse out, cells out, check VBAT to GND with continuity (must not beep).

STAGE 14 45 min

Solar panel and charger (CN3791)

Connect the panel to the charger, and the charger to the buses.

PARTS FOR THIS STAGE

- Solar panel
- CN3791 and its 2 plug cables
- heat-shrink
- electrical tape

TOOLS

- Iron
- cutter
- stripper
- multimeter

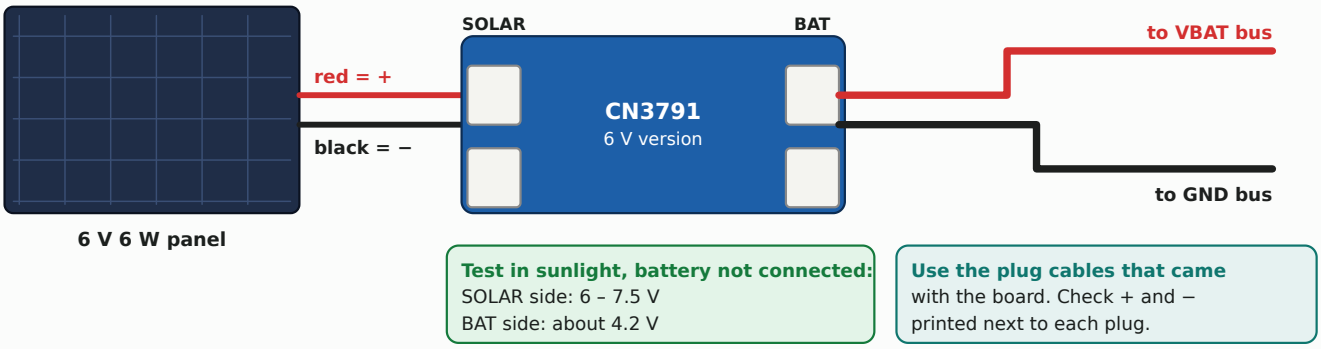


Figure 18. Solar panel → CN3791 → battery bus. Either SOLAR plug works (they are connected together), and the same for BAT.

1. Lay the panel **face down** on cloth so it makes no voltage while you work.

2. Cut the panel's plug off, leaving as much cable as possible. Split the two wires apart for 5 cm, strip and tin them.

3. Find which panel wire is +: turn the panel face up in the light, red probe on one wire, black on the other. A positive number means the red probe is on +. Mark it with red tape. Turn the panel face down again.

4. Take the plug cables that came with the CN3791. Plug one into a **SOLAR** socket. Look at the board next to the socket for the printed + and −, and see which cable wire is on +.

5 Unplug it. Slide heat-shrink onto the cable wires. Solder panel + to the cable's + wire and panel − to its − wire. Shrink, then wrap the joint in electrical tape.

6 Plug a second cable into a **BAT** socket, check + and − the same way, unplug. Solder its + wire to the VBAT bus at **tap 2** and − wire to the GND bus at **tap B**.

7 Plug both cables back in: panel cable in SOLAR, bus cable in BAT.

8 Put the cells in and the fuse in. Turn the panel face up in direct sunlight.

CHECK

MEASURE BETWEEN	MULTIMETER ON	EXPECTED
CN3791 SOLAR + and −	V⎓ 20	**5 – 7.5 V in sun**
VBAT bus and GND bus	V⎓ 20	**slowly rises over an hour; never above 4.2 V**

One LED on the CN3791 lights while charging. If the BMS was asleep in Stage 13, it is awake now.

BALANCING UNEQUAL CELLS (ONLY IF STAGE 10 TOLD YOU TO)

Put only one cell in the holder and leave the panel in the sun until that cell reads 4.15 – 4.20 V. Repeat for the other two. Then insert all three together.

IF THE CHECK FAILS

- SOLAR side reads 0 V: panel wires swapped or the plug not pushed in, or no direct sun.
- BAT side reads nothing and the LEDs are off: + and − of the BAT cable are swapped. Fuse out, fix it.
- Voltage does not rise: weak sun (clouds, window glass blocks a lot), or the battery is already full.

STAGE 15 20 min

3.3 V converter (TPS63020)

Makes a steady 3.3 V for the ESP32, whether the battery is full or nearly empty.

PARTS FOR THIS STAGE

- TPS63020
- 22 AWG red and black
- orange tag

TOOLS

- Iron
- multimeter

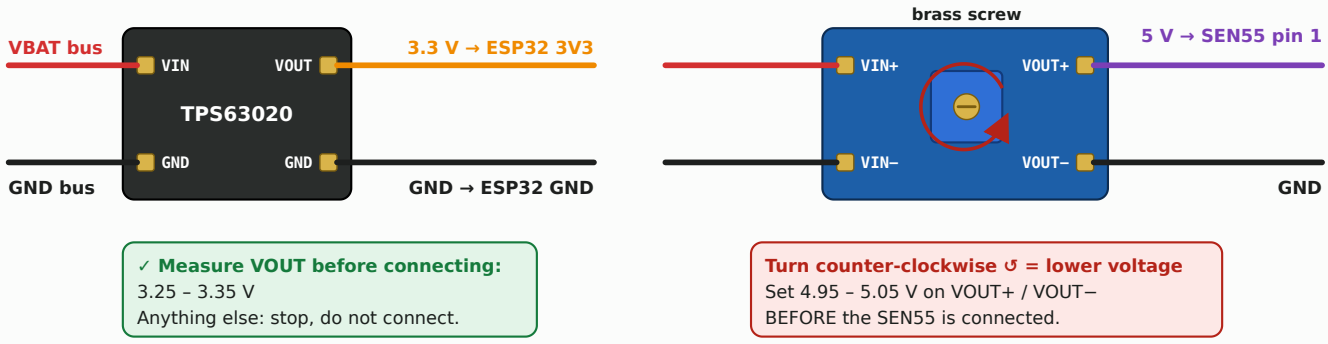


Figure 19. The two voltage converters. Both take power from the VBAT bus. Measure their outputs before anything is connected to them.

1. Fuse out.

2. Look at the TPS63020 board: find VIN and GND on one side, VOUT and GND on the other (they may be printed on the back).

3. Solder a 10 cm red 22 AWG wire from **VIN** to VBAT bus **tap 3**. Solder a 10 cm black wire from its input-side **GND** to GND bus **tap C**.

4. Solder a 20 cm red 22 AWG wire to **VOUT**. Put an **orange** heat-shrink tag on both ends: this is the 3.3 V wire. Solder a 20 cm black wire to the output-side **GND**.

5. Tape the free ends of these two output wires separately.

6. Fuse in. Measure the output wires.

CHECK

MEASURE BETWEEN	MULTIMETER ON	EXPECTED
TPS63020 VOUT wire and its GND wire	V⎓ 20	**3.25 – 3.35 V**

IF THE CHECK FAILS

- Reads about 5 V: it is the 5 V version. Do not connect it to the ESP32. Contact the seller.
- Reads 0 V: VIN and GND swapped, or the bus wires not soldered. Fuse out, check with continuity.

STAGE 16 20 min

5 V converter (MT3608)

Makes 5 V for the SEN55. It arrives set to a random voltage, often far too high.

PARTS FOR THIS STAGE

- MT3608
- 22 AWG red and black
- tag
- nail polish or hot glue

TOOLS

- Iron
- small flat screwdriver
- multimeter

1. Fuse out.

2. Solder a 10 cm red 22 AWG wire from **VIN+** to VBAT bus **tap 4**, and a black wire from **VIN−** to GND bus **tap D**.

3. Solder a 20 cm red wire to **VOUT+** with a **purple** (or any other unused colour) tag at both ends: this is the 5 V wire. Solder a 20 cm black wire to **VOUT−**. Tape the free ends.

4. Fuse in. Multimeter on V⎓ 20 on VOUT+ and VOUT−. It probably reads 6 – 28 V.

5. With a small flat screwdriver, turn the brass screw on the blue box **counter-clockwise**. Turn slowly, a full turn at a time. The voltage drops. It can take 10 – 25 turns before anything changes: that is normal.

6. Stop at **5.0 V** (4.95 – 5.05). If you went too low, turn clockwise a little.

7. Fuse out, fuse in, measure again: still 5.0 V.

8. Put a drop of nail polish or hot glue on the brass screw so vibration cannot turn it.

CHECK

MEASURE BETWEEN	MULTIMETER ON	EXPECTED
MT3608 VOUT+ and VOUT−	V⎓ 20	**4.95 – 5.05 V**

NEVER SKIP THIS

The SEN55 accepts at most 5.5 V. Connecting it to an unadjusted MT3608 destroys it instantly.

IF THE CHECK FAILS

- Voltage does not change at all after 30 turns: turn the other way 30 turns; some boards are reversed.
- Reads about the battery voltage (3.7 V) and cannot go higher: input wires swapped or the board is faulty.

PART 3

Build the control board

The ESP32 board, the sensor and the modem, each tested with a small program before the real one.

STAGE 17 1 – 2 h

Make the ESP32 board

Sockets for the ESP32, two power rails, and two pull-up resistors.

PARTS FOR THIS STAGE

- 7 × 9 cm perfboard
- female headers
- 2 × 4.7 kΩ
- wire

TOOLS

- Iron
- cutter
- pliers
- multimeter

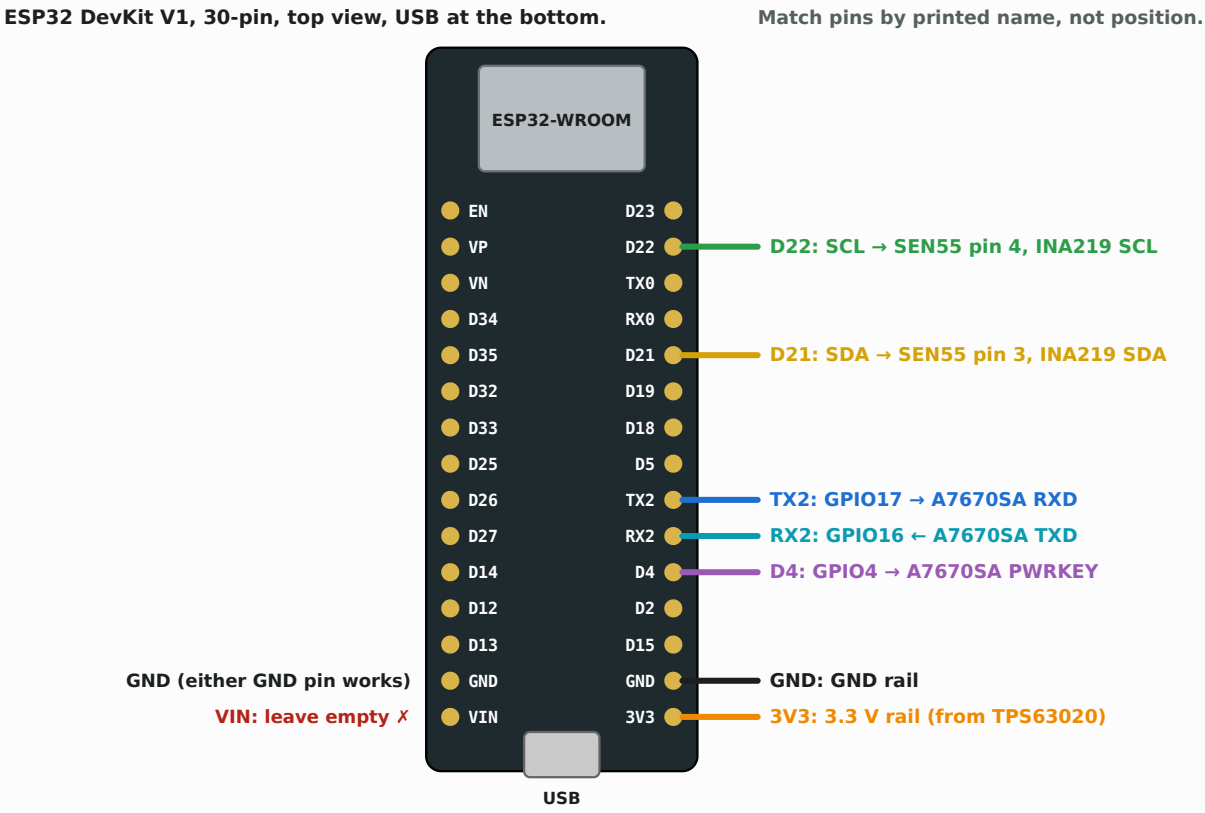


Figure 20. The ESP32 pins this project uses, all on one side of a 30-pin DevKit V1. **TX2 = GPIO17** and **RX2 = GPIO16** on the board's labels.

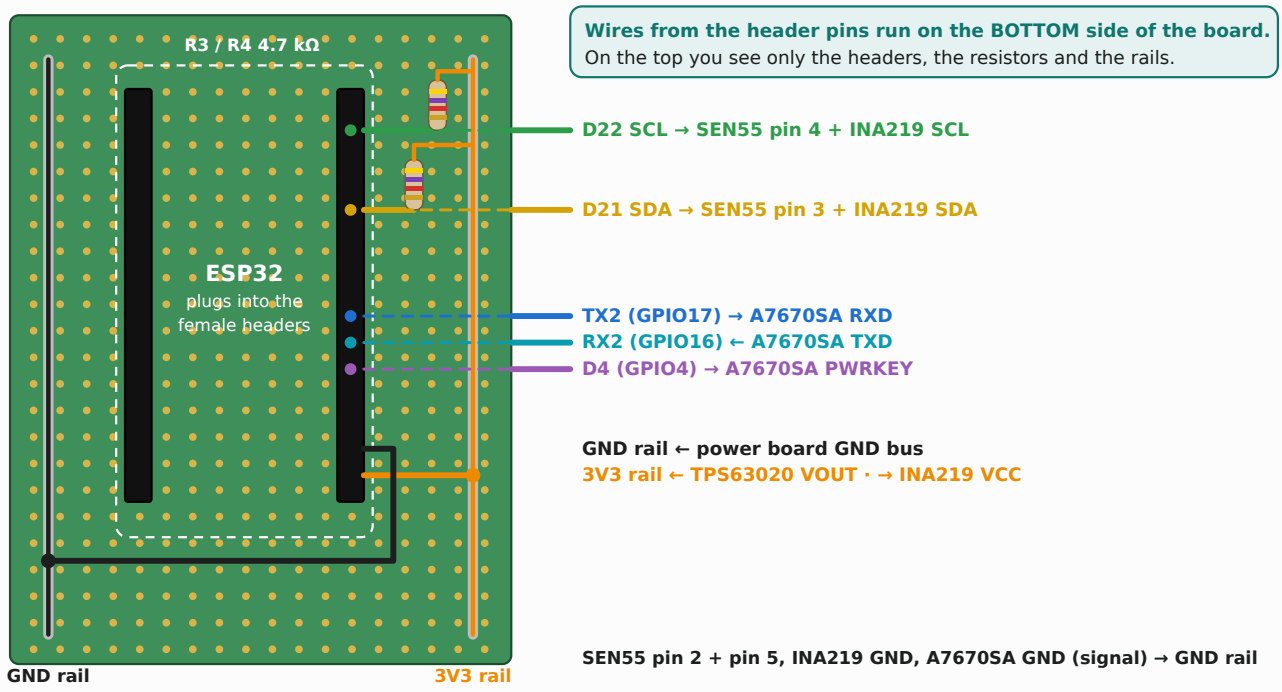


Figure 21. The ESP32 board (7 × 9 cm). Two rails (GND and 3V3), two pull-up resistors, and solder points for the sensor and modem wires.

1. Count the pins on one side of your ESP32 (15 on the usual 30-pin board). Cut two female header pieces of that length (Figure 12).

2. **Push both headers onto the ESP32's pins**. Set the ESP32 with its headers onto the 7 × 9 cm board, roughly centred, USB end at one short edge. This sets the exact spacing.

3. Turn the board over carefully, holding the ESP32. Solder one pin at each end of each header. Turn it over and check the headers sit flat and straight. Then solder all the other pins.

4. Pull the ESP32 out of the headers and put it back in its bag.

5. Make two bus wires (18 or 22 AWG, bare, tinned) along the long edges, as in Figure 16: the left one is the **GND rail**, the right one the **3V3 rail**. Mark them.

6 On the bottom, with a short wire, connect the header pin under the ESP32's **3V3** pin to the 3V3 rail, and the pin under **GND** to the GND rail (Figure 20 shows which pins).

7 Find the header pins under **D21** and **D22**. Solder resistor R3 (4.7 kΩ) from the D21 pin to the 3V3 rail, and R4 from the D22 pin to the 3V3 rail (through-hole method).

8 Bring the TPS63020 output wires over: the orange-tagged 3.3 V wire to the 3V3 rail, its black wire to the GND rail.

9 Cut 20 cm of black 18 AWG. Solder one end to the ESP32 board GND rail and the other to the power board GND bus at **tap F**.

CHECK: ESP32 NOT PLUGGED IN, FUSE IN

MEASURE BETWEEN	MULTIMETER ON	EXPECTED
3V3 rail and GND rail	V⎓ 20	**3.25 – 3.35 V**
Header pin under D21 and 3V3 rail (fuse out)	Ω 20k	**about 4.7**
Header pin under VIN and anything else (fuse out)	continuity	**no beep: VIN stays empty**

IF THE CHECK FAILS

- Headers crooked: reheat one pin at a time while pressing the header straight, or remove with the pump and redo.
- 3V3 rail reads 0 V: the TPS63020 output wire or the GND link is not soldered; check with continuity.

STAGE 18 45 min

Install the Arduino IDE and drivers

The free program used to write and upload code to the ESP32. Done once per computer.

PARTS FOR THIS STAGE

- ESP32
- USB data cable

TOOLS

- Laptop with internet



Figure 22. Adding the ESP32 boards address in Preferences. You do this once per computer.

1. On the team laptop, open a browser and go to **arduino.cc/en/software**. Download **Arduino IDE 2** for Windows (the "Win 10 and newer, 64 bits" installer) and run it. Accept all defaults. Allow it to install drivers if Windows asks.

2. Open Arduino IDE. Click **File → Preferences**. In "Additional boards manager URLs" paste the address below, then click OK.
   `https://espressif.github.io/arduino-esp32/package_esp32_index.json`

3. Click the second icon on the left side bar (a board icon), or **Tools → Board → Boards Manager**. Type **esp32** in the search. Find **esp32 by Espressif Systems** and click **Install**. It is large (several hundred MB): wait until it says INSTALLED.

4 Click the third icon on the left (books), or **Tools → Manage Libraries**. Search and install each of these. When asked "install dependencies too?", click **Install all**:

5 **Sensirion I2C SEN5X**

(by Sensirion)

6 **Adafruit INA219**

(by Adafruit)

7 **ArduinoJson**

(by Benoit Blanchon), version 7.x

8 Plug the ESP32 (on its own, out of the board) into the laptop with a USB cable. Open Device Manager and look under "Ports (COM & LPT)" (Figure 23).

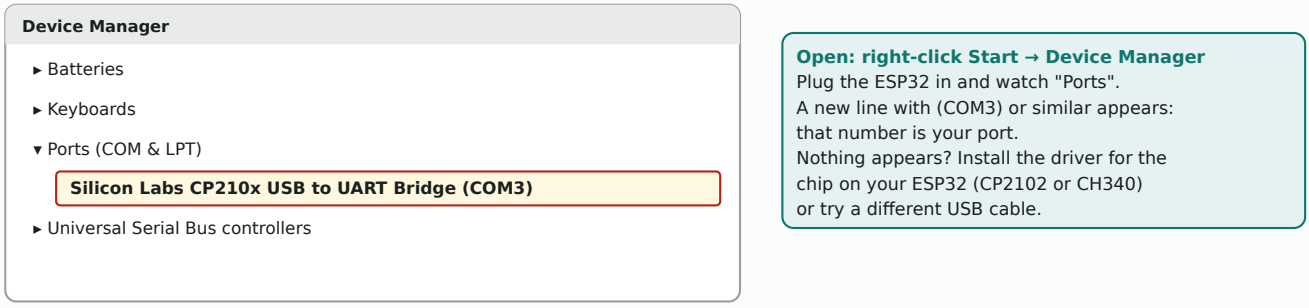


Figure 23. Finding the ESP32's COM port on Windows. The name says CP210x or CH340 depending on the USB chip on your board.

NO NEW COM PORT APPEARS

- Many USB cables only charge and cannot carry data. Try another cable (one that transfers files from a phone works).
- Look at the small chip next to the ESP32's USB socket. It says **CP2102** or **CH340**. Search the web for "CP210x driver Silicon Labs" or "CH340 driver WCH", install it from the maker's site, restart the laptop, plug in again.
- Try another USB port on the laptop.

STAGE 19 20 min

Your first program: Blink

Proves the laptop, cable and ESP32 all work, before anything else is connected.

PARTS FOR THIS STAGE

- ESP32
- USB cable

TOOLS

- Laptop

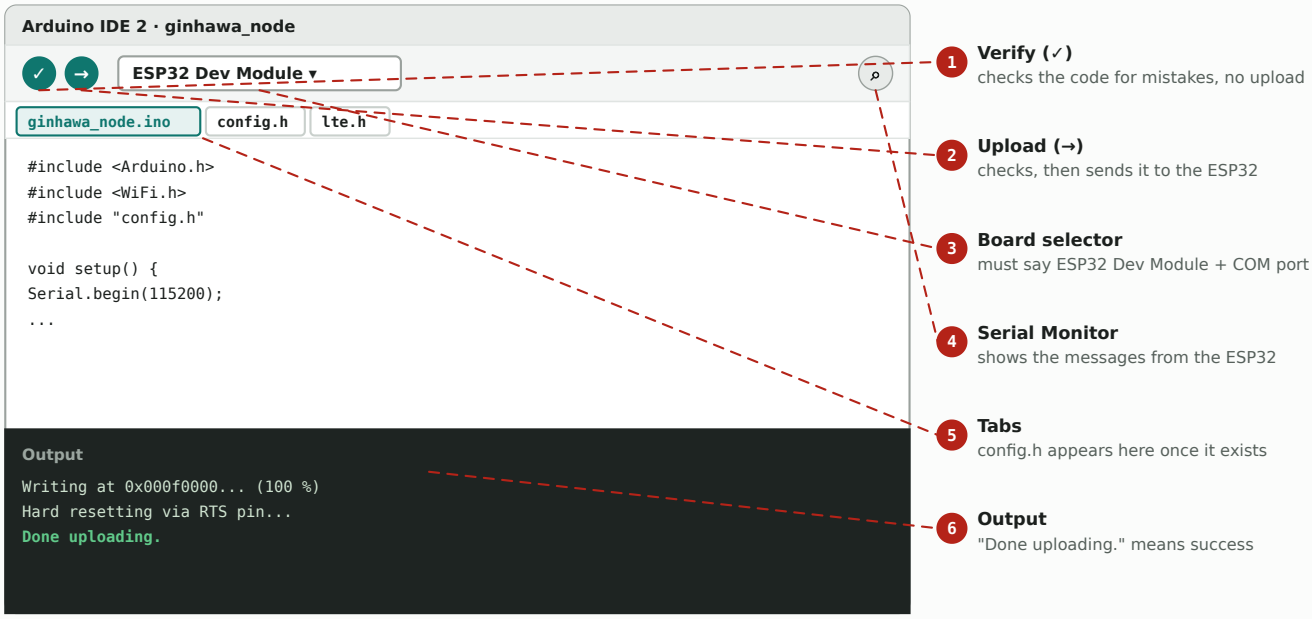


Figure 24. The Arduino IDE 2 window and the six things you will use. (Simplified drawing; colours on your screen may differ.)

1. In Arduino IDE: **File → New Sketch**. Delete everything in the window.

2. Copy the Blink program below (Copy button) and paste it into the window.

3. At the top, click the board selector and choose **Select other board and port**. Type **ESP32 Dev Module**, select it, and select your COM port. Click OK.

4. Click **Upload** (the → arrow). The first time it asks where to save: make a folder called "Blink". Compiling takes a minute the first time.

5 Watch the Output pane. When it shows "Connecting....", if it does not continue within 5 seconds, press and hold the ESP32's **BOOT** button until percentages appear, then release.

6 When it says **Done uploading**, look at the ESP32.

Blink

```
// Blink: the "hello world" of electronics. Flashes the ESP32's blue LED.
#define LED_BUILTIN 2

void setup() {
  pinMode(LED_BUILTIN, OUTPUT);
}

void loop() {
  digitalWrite(LED_BUILTIN, HIGH);  // LED on
  delay(1000);                      // wait 1 second
  digitalWrite(LED_BUILTIN, LOW);   // LED off
  delay(1000);
}
```

CHECK

The small blue LED on the ESP32 blinks: 1 second on, 1 second off. Your setup works.

IF THE CHECK FAILS

- "Port not found" or greyed out: COM port not selected, or another program (Serial Monitor in another window) is using it.
- "Failed to connect to ESP32": hold BOOT during "Connecting....". Try a slower upload: Tools → Upload Speed → 115200.
- No blue LED: some boards have no LED on pin 2. If upload said "Done uploading", it worked: continue.

STAGE 20 1 – 2 h

Connect the SEN55 and INA219, then test

Both talk to the ESP32 over the same two I²C wires.

PARTS FOR THIS STAGE

- SEN55
- JST GH cable
- INA219 (already on the power path)
- 22 AWG wire
- tags

TOOLS

- Iron
- stripper
- multimeter
- laptop

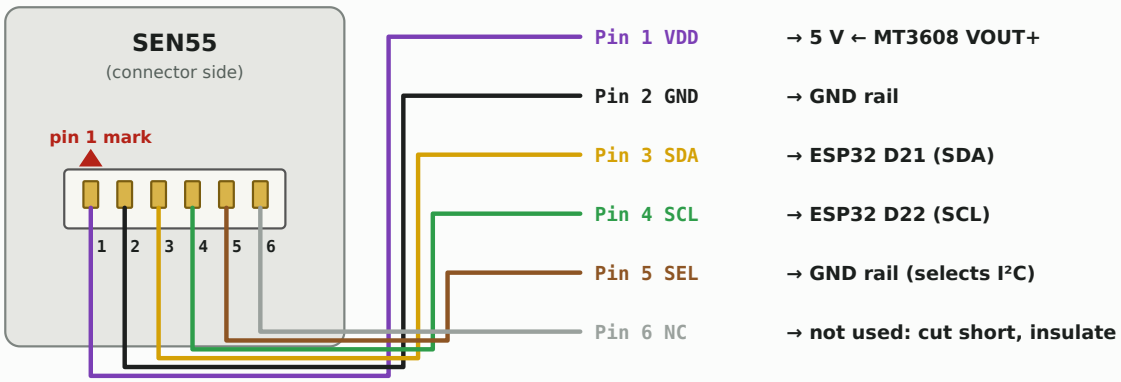


Figure 26. SEN55 pin-out and where each wire goes. Find **pin 1** from the mark printed next to the sensor's connector, then count. Go by pin number, never by wire colour.

1. Fuse out.

2. Take one JST GH cable. Cut it exactly in the middle. Strip 4 mm of each of the 6 wires on one half; tin them. Keep the other half as a spare.

3. Plug the cable gently into the SEN55. Find the **pin 1** mark near the SEN55 socket (a small triangle or "1" on the label).

4. With the multimeter on continuity, put one probe on the bare end of a wire and touch the other probe to the metal contact in the plug at the pin 1 position (use a needle or thin wire if needed). Find which wire is pin 1. Repeat for pins 2 – 6. **Write down: pin 1 = (colour), pin 2 = (colour) …**

5 Solder **pin 1 (VDD)** to the purple-tagged 5 V wire from the MT3608 VOUT+. Cover with heat-shrink.

6 Solder **pin 2 (GND)** and **pin 5 (SEL)** to the ESP32 board GND rail. Solder the MT3608 VOUT− black wire to the GND rail too.

7 Solder **pin 3 (SDA)** to the header pin under **D21**, and **pin 4 (SCL)** to the header pin under **D22** (on the bottom of the board). Tag them yellow and green.

8 Cut **pin 6** short and cover its end with heat-shrink.

9 INA219 small pins, with four 22 AWG wires: **VCC** → 3V3 rail, **GND** → GND rail, **SDA** → D21 pin, **SCL** → D22 pin.

10 Plug the ESP32 into its headers (check every pin enters a socket, none outside). Fuse in. Then connect the USB cable to the laptop.

11 In Arduino IDE: File → New Sketch, paste Test 1 below, Upload. Then click the Serial Monitor (magnifier, top right) and set it to **115200 baud** (Figure 25).

Test 1: I²C scanner

```
// Ginhawa test 1: finds devices on the I2C bus (SDA = D21, SCL = D22)
#include <Wire.h>

void setup() {
  Serial.begin(115200);
  delay(1000);
  Wire.begin();  // SDA 21, SCL 22
}

void loop() {
  Serial.println("Scanning...");
  int found = 0;
  for (byte addr = 1; addr < 127; addr++) {
    Wire.beginTransmission(addr);
    if (Wire.endTransmission() == 0) {
      Serial.printf("  found 0x%02X", addr);
      if (addr == 0x40) Serial.print("  <- INA219 (battery meter)");
      if (addr == 0x69) Serial.print("  <- SEN55 (air sensor)");
      Serial.println();
      found++;
    }
  }
  if (found == 0) Serial.println("  nothing found: check SDA/SCL, power and GND");
  Serial.println();
  delay(3000);
}
```

What a good result looks like (repeats every 3 seconds)

```
Scanning...
  found 0x40  <- INA219 (battery meter)
  found 0x69  <- SEN55 (air sensor)
```

USB AND BATTERY TOGETHER

Having USB connected while the fuse is in is fine for short tests. When the station is finished, unplug USB for good.

IF THE CHECK FAILS

- Nothing found: SDA and SCL swapped (swap the two wires at D21/D22), no common GND (check the tap F link), or 3V3 rail off.
- Only 0x40 (INA219): SEN55 not powered or pin order wrong. Measure 5 V between pin 1 and pin 2 wires at the sensor end.
- Only 0x69 (SEN55): INA219 VCC or its SDA/SCL wires.
- Garbage characters in the Serial Monitor: the speed is not 115200.

STAGE 21 1 – 2 h

Connect the 4G modem (A7670SA) and test

Power, two data wires, a power-key wire, antennas and the SIM.

PARTS FOR THIS STAGE

- A7670SA
- antennas
- nano-SIM with load
- 2 × 1000 µF
- 18 and 22 AWG

TOOLS

- Iron
- multimeter
- laptop

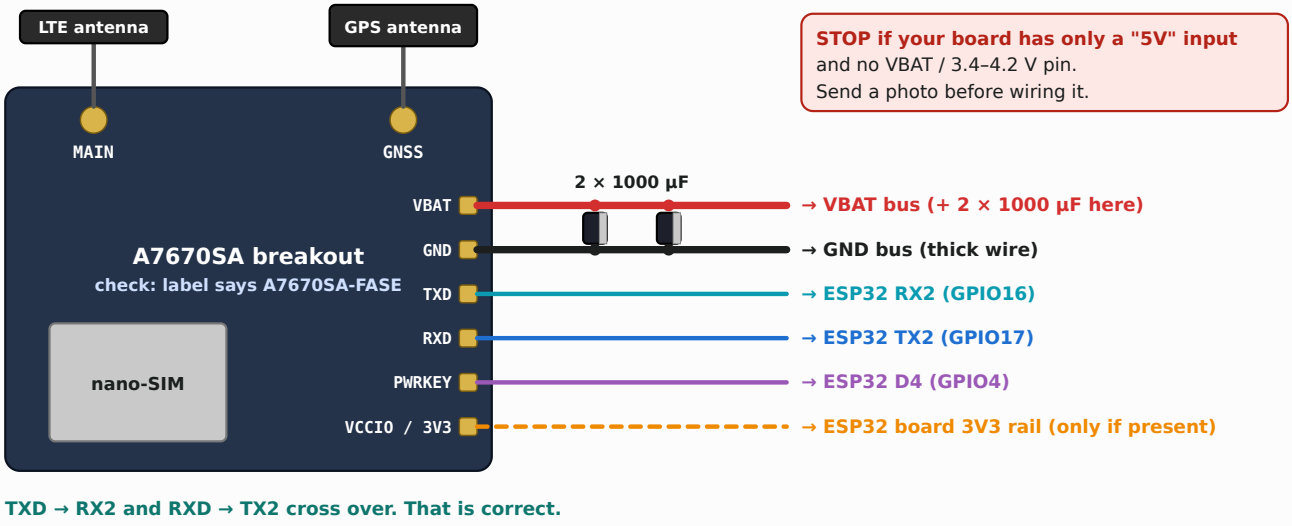


Figure 27. A7670SA connections. Pin names differ between breakout makers: PWRKEY may be printed as **PWR**, **PEN** or **K**; VBAT may be **VCC 3.4–4.2 V** or **BAT**.

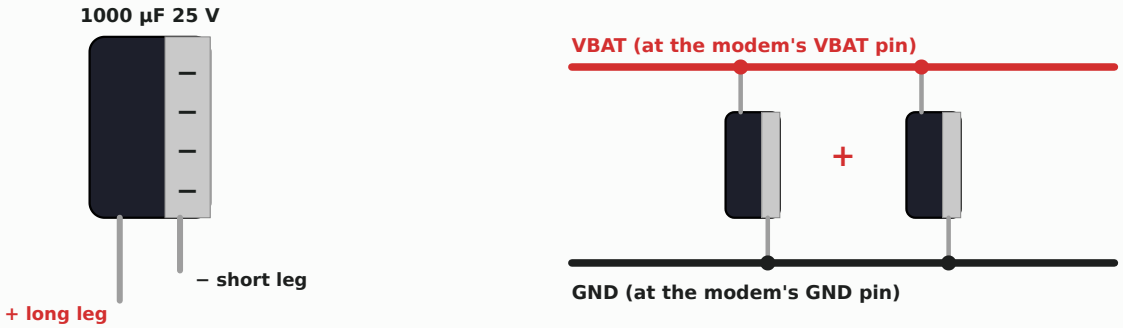


Figure 28. Capacitor polarity. The side with the stripe is **−**. Two capacitors side by side, both + to VBAT and both − to GND.

1. Compare your board with Figure 27. Find and write down where these are printed: **VBAT** (or VCC 3.4 – 4.2 V, BAT), **GND**, **TXD**, **RXD**, **PWRKEY** (or PWR, PEN, K). Some boards also have **VCCIO** or **3V3**.

2. If your board has only a **5V** power input and no VBAT pin: **stop**. Send a clear photo of both sides before wiring it.

3. Fuse out. Cut 15 cm of red and 15 cm of black 18 AWG. Solder red from VBAT bus **tap 5** to the modem **VBAT**; black from GND bus **tap E** to the modem **GND**. Keep these wires short and straight.

4. Solder the two 1000 µF capacitors right at the modem: each long leg (+) to the VBAT pin or wire, each striped side (−) to GND. Cover the legs with heat-shrink.

5. Signal wires (22 AWG, about 15 cm): modem **TXD** → ESP32 board **RX2** pin (GPIO16), modem **RXD** → **TX2** pin (GPIO17), modem **PWRKEY** → **D4** pin. Tag them blue. If the board has VCCIO/3V3: one more wire to the 3V3 rail. Also one thin black wire from a modem GND pin to the ESP32 board GND rail.

6. Antennas: press the 4G antenna plug onto **MAIN** (or LTE) and the GPS antenna onto **GNSS** (or GPS), straight down until they snap (Figure 13).

7. Insert the registered nano-SIM with data load into the slot, cut corner the way the slot drawing shows.

8. Fuse in, USB in. Upload Test 2. Open the Serial Monitor: 115200 baud and **Both NL & CR**. Wait 15 seconds, then type each command in the table and press Enter.

Test 2: modem terminal

```
// Ginhawa test 2: talk to the A7670SA. Type AT commands in the Serial Monitor.
// Serial Monitor: 115200 baud, "Both NL & CR".
#define MODEM_RX 16   // ESP32 RX2 <- modem TXD
#define MODEM_TX 17   // ESP32 TX2 -> modem RXD
#define PWRKEY    4   // ESP32 D4  -> modem PWRKEY

void setup() {
  Serial.begin(115200);
  Serial2.begin(115200, SERIAL_8N1, MODEM_RX, MODEM_TX);
  pinMode(PWRKEY, OUTPUT);
  Serial.println("Pressing the modem's power key for 1 second...");
  digitalWrite(PWRKEY, HIGH);
  delay(1000);
  digitalWrite(PWRKEY, LOW);
  Serial.println("Wait about 15 seconds, then type: AT");
}

void loop() {
  while (Serial.available())  Serial2.write(Serial.read());   // you -> modem
  while (Serial2.available()) Serial.write(Serial2.read());   // modem -> you
}
```

TYPE	GOOD ANSWER	MEANING
AT	OK	The modem is on and listening
AT+SIMCOMATI	Model: A7670SA...	Model and firmware
AT+CPIN?	+CPIN: READY	The SIM is detected and has no PIN
AT+CSQ	+CSQ: 18,99	Signal: first number 10 – 31 is usable. 99,99 = no signal
AT+CEREG?	+CEREG: 0,1	Registered on 4G (0,5 = roaming, also fine)
AT+CGNSSPWR=1	+CGNSSPWR: READY!	GPS switched on
AT+CGPSINFO	+CGPSINFO: 1429.1234,N,12100.5678,E,...	A position. Empty commas = no fix yet: take it outside, wait 1 – 5 min, repeat
AT+CGNSSPWR=0	OK	GPS off again

Example of a good session

```
Pressing the modem's power key for 1 second...
Wait about 15 seconds, then type: AT
AT
OK
AT+CPIN?
+CPIN: READY
OK
AT+CSQ
+CSQ: 18,99
OK
AT+CEREG?
+CEREG: 0,1
OK
```

IF THE CHECK FAILS

- No answer to AT: swap the TXD/RXD wires; check VBAT at the modem pins (3.6 – 4.2 V); press the board's own power button if it has one.
- Still nothing: the board's PWRKEY may need the opposite signal. Note what its LEDs do and report it; the firmware can be adjusted.
- +CPIN: SIM not inserted or ERROR: SIM upside down, not registered, or has a PIN (remove it using a phone).
- +CSQ: 99,99 for minutes: antenna on the wrong socket, or no coverage indoors: try near a window or outside.
- Modem restarts when it connects: battery low, or VBAT wires too long/thin. Charge the battery; add the third capacitor.

PART 4

Software and the app

Run the Ginhawa app on the laptop, put the real program on the station, and test it for a day on the table.

STAGE 22 30 min

Start the Ginhawa app on the laptop

The station needs somewhere to send its readings: the app running in Docker on your laptop.

PARTS FOR THIS STAGE

- Laptop with Docker Desktop and the ginhawa folder

TOOLS

- PowerShell

1. Make sure **Docker Desktop** is installed and running (whale icon in the taskbar, says "Engine running").

2. Open **PowerShell** and go to the project folder: `cd C:\Users\peter\ginhawa`

3. Find the laptop's Wi-Fi address: type `ipconfig` and press Enter. Under "Wireless LAN adapter Wi-Fi", copy the **IPv4 Address** (example 192.168.1.20). Write it down.

4. Open the `.env` file in Notepad (`notepad .env`). Set these two lines with your address, save, close:
   `SITE_ADDRESS=localhost, 192.168.1.20`
   `DEFAULT_SNI=192.168.1.20`

5. In the same file, copy the value after `ENROLL_SECRET=`. You need it in the next stage. Do not share it outside the team.

6. Start the app: `docker compose up -d --build`. Wait until it finishes (first time: several minutes).

7 On the laptop browser open **https://localhost**. Accept the certificate warning (Advanced → Proceed). The Ginhawa app opens.

KEEP THE IP ADDRESS FIXED

The laptop's address can change when it reconnects to Wi-Fi. If the station later says "Upload failed (HTTP -1)", run `ipconfig` again and compare. A router setting called "DHCP reservation" keeps it fixed.

STAGE 23 30 min

Upload the Ginhawa firmware

The real program: it reads the sensor every second, uploads once a minute, and adds the station to the app by itself.

1. Open `C:\Users\peter\ginhawa\firmware\ginhawa_node` in File Explorer.

2. Copy `config.example.h` and paste it in the same folder. Rename the copy to exactly `config.h` (Windows may hide ".h": turn on View → File name extensions).

3. Double-click `ginhawa_node.ino`. Arduino IDE opens it with tabs for config.h and lte.h.

4. Click the **config.h** tab and change the lines shown below. Keep the quote marks. Save (Ctrl+S).

5. Select **ESP32 Dev Module** and your COM port as in Stage 19. Click **Upload**.

6. Open the Serial Monitor at 115200 baud and watch.

config.h: the lines you change

```
#define WIFI_SSID      "YourWiFiName"
#define WIFI_PASSWORD  "YourWiFiPassword"
#define API_BASE_URL   "https://192.168.1.20"      // your laptop's IP address (see
below)
#define ENROLL_SECRET  "same-as-ENROLL_SECRET-in-.env"
#define LTE_APN        "internet.dito.ph"          // DITO. Smart: "internet"  Globe:
"internet.globe.com.ph"
```

LINE	WHAT TO PUT
WIFI_SSID	The Wi-Fi name exactly, capitals matter. Must be 2.4 GHz (the ESP32 cannot see 5 GHz networks).
WIFI_PASSWORD	The Wi-Fi password.
API_BASE_URL	"https://" + the laptop address from Stage 22. No slash at the end.
ENROLL_SECRET	Exactly the value from .env.
LTE_APN	DITO: internet.dito.ph · Smart: internet · Globe: internet.globe.com.ph
STATION_NAME	Leave "" to be named after the barangay automatically.

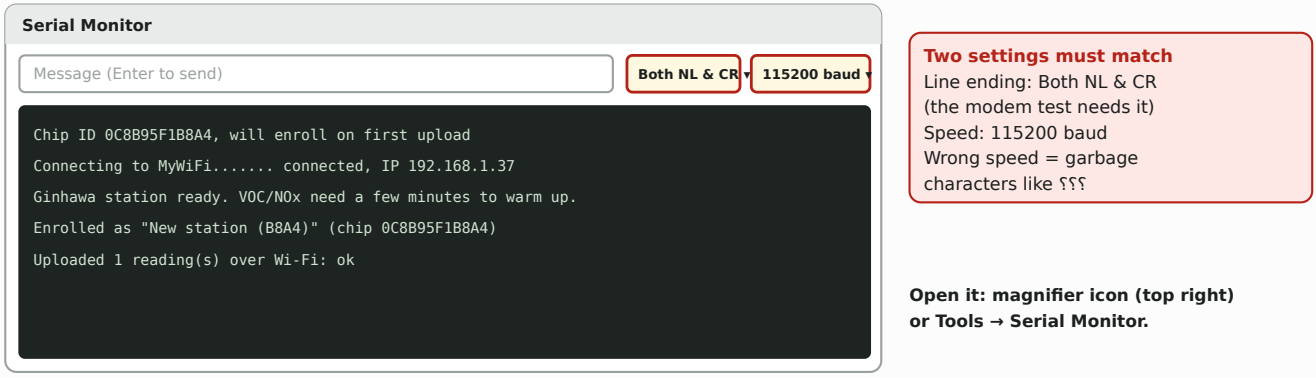


Figure 25. The Serial Monitor shows what the ESP32 is doing. The text is an example of a healthy first start.

CHECK: IN ORDER, WITHIN 2 MINUTES

1. Chip ID … will enroll on first upload
2. Connecting to YourWiFi….. connected, IP …
3. Ginhawa station ready.
4. Enrolled as "New station (…)"
5. Uploaded 1 reading(s) over Wi-Fi

Then open the app: a new station card says **Just added**.

IF THE CHECK FAILS

- "Connecting … failed": SSID/password typo, or a 5 GHz network. Use a phone hotspot set to 2.4 GHz to test.
- "Enrollment failed (HTTP 401/403)": ENROLL_SECRET does not match .env exactly, or the device was blocked in the app.
- "Upload failed (HTTP -1)": the laptop address is wrong, Docker is not running, or the laptop is on a different Wi-Fi.
- Compile error "config.h: No such file": the file is named config.h.txt; turn on file extensions and rename.
- Compile error about a library: install the three libraries from Stage 18.

STAGE 24 1 day

24-hour bench test

Run the complete station on the table for a day before it goes into the box.

1. Unplug USB. The station now runs only from the battery, exactly as it will outdoors.
2. Put the solar panel by a sunny window or outside, with the electronics indoors on the table.
3. Check the app every few hours: readings arrive every minute; PM values change every 5 minutes; battery level is shown.
4. Breathe near the sensor or light a match nearby (then blow it out) and watch VOC and PM rise, then fall.
5. Feel the converters, modem and fuse holder: warm is fine, too hot to hold is not.
6. After 24 hours write down: battery voltage, highest temperature reading, any gaps in the data. Keep this for the paper.

TESTING THE 4G FALLBACK

Over 4G the station can only reach a server on the internet, not the laptop at home. To test 4G properly, the app must be online (domain or VPS). Until then, check with Test 2 that the modem registers and gets signal.

IF THE CHECK FAILS

- Data stops after some hours: check the battery voltage; if low, the panel is not charging (see Stage 14 checks).
- Readings stop and the ESP32 restarts often: a loose wire; press gently on joints while watching the Serial Monitor.

PART 5

Box, mounting and running

Put the tested electronics in a waterproof box, mount it outdoors, and keep it running.

STAGE 25 1 – 2 h

Prepare the enclosure

Drill the holes before anything goes inside.

PARTS FOR THIS STAGE

- IP65 box
- PG7 gland
- M12 vent
- mesh
- standoffs, screws

TOOLS

- Drill, 3 and 12 mm bits or step bit
- cutter
- file
- masking tape, marker
- hot glue

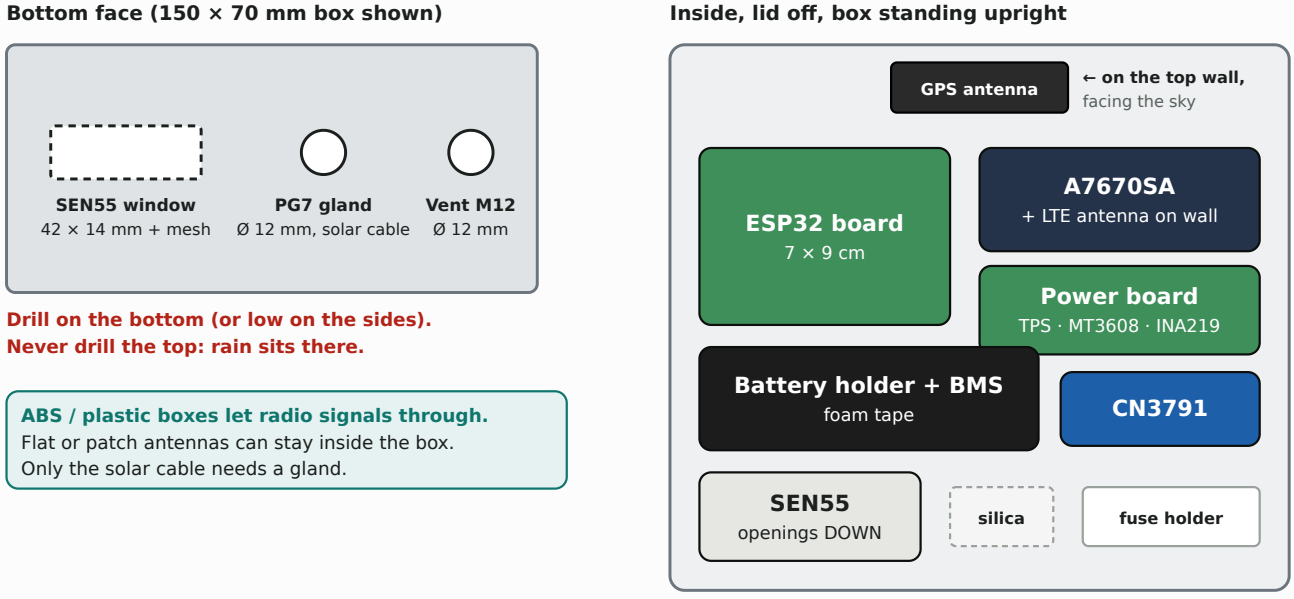


Figure 30. Drilling plan and layout inside an IP65 box of about 150 × 100 × 70 mm. Adjust the positions to the box you buy, keeping these rules.

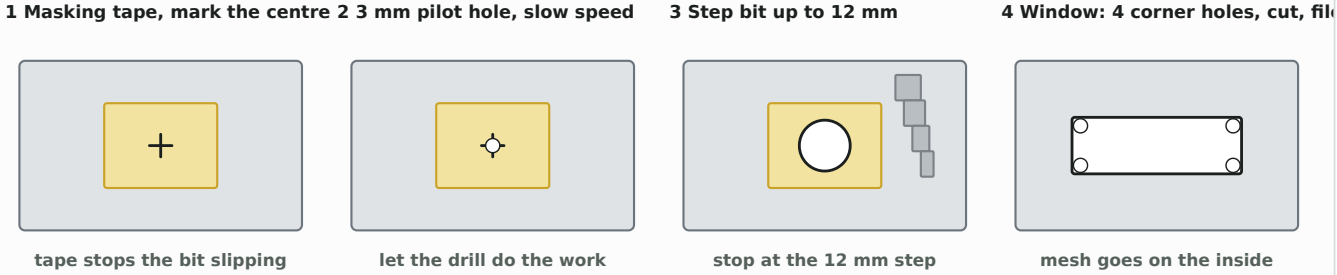


Figure 29. Drilling the plastic box. Go slowly: plastic cracks if you push hard or drill fast.

1. Buy an **IP65 ABS junction box**, about 150 × 100 × 70 mm or bigger, with mounting ears or a mounting plate.

2. Lay every part inside with the lid on to check it closes. Plan where each goes as in Figure 30.

3. Stick masking tape on the bottom face. Mark: one 12 mm hole for the gland, one 12 mm hole for the vent, and the SEN55 window (42 × 14 mm). Keep 15 mm from the edges and between holes.

4. Clamp or hold the box firmly on a table. Drill a 3 mm pilot hole at each centre, slowly.

5. Enlarge to 12 mm with the step bit (stop at the 12 mm step) or a 12 mm bit at low speed.

6. Window: drill 3 mm holes in the four corners, then cut the lines between them with a cutter (many light cuts, not one hard cut). Smooth with a file.

7. Cut the stainless mesh 10 mm bigger than the window on all sides. Glue it on the **inside** with hot glue or epoxy all around, no gaps.

8. Fit the gland: body through from outside, rubber seal on, nut tight inside (hand-tight plus a quarter turn). Fit the vent the same way, its O-ring on the outside.

9 Hold the ESP32 board against the back wall where it will go. Mark its four corner holes through the board, drill 3 mm.

10 Remove the masking tape and all plastic shavings.

IF THE CHECK FAILS

- Plastic cracked: seal the crack inside with epoxy; if near a hole, use another position.

STAGE 26 1 – 2 h

Put everything in the box

Fix every part so nothing moves or touches when the box shakes on a pole.

PARTS FOR THIS STAGE

- all tested electronics
- foam tape
- zip ties
- silica gel

TOOLS

- Allen key
- pliers
- laptop

1. Fuse out, cells out.

2. Screw the four standoffs to the back wall: M3 × 8 screw from outside with the 2 mm Allen key into each standoff. Put the ESP32 board on the standoff threads and fix with M3 nuts.

3. With foam tape, stick the battery holder, power board, CN3791 and modem board to the box walls. Put a zip tie or tape strip around the holder so cells cannot fall out.

4. Mount the SEN55 with its two air openings facing **down** against the mesh window, foam tape on its back. Keep it at least 3 cm from the modem and converters (they are warm).

5. GPS antenna: stick it to the inside of the **top** wall, flat side toward the sky. 4G antenna: flat on a side wall, away from the GPS antenna.

6. Solar cable: pass it through the gland from outside, leave a U-shaped drip loop outside, and tighten the gland until the cable cannot be pulled.

7. Tie loose wires into bundles with small zip ties. No bare metal touches the box, another board or the mesh.

8. Cells in, fuse in. Connect USB briefly and watch the Serial Monitor for "Uploaded … over Wi-Fi". Unplug USB.

9 Put 1 – 2 silica gel packs inside. Close the lid evenly, tightening the screws a little at a time in a cross pattern so the seal presses all round.

CHECK BEFORE CLOSING

- Gently shake the box: nothing rattles or moves.
- Every joint covered; no tape coming loose.
- A new reading appears in the app after closing.

STAGE 27 1 h

Mount it outside

Where the sensor measures the air people breathe, with sun on the panel.

PARTS FOR THIS STAGE

- hose clamps or U-bolts
- panel bracket
- zip ties

TOOLS

- Screwdriver, wrench, ladder

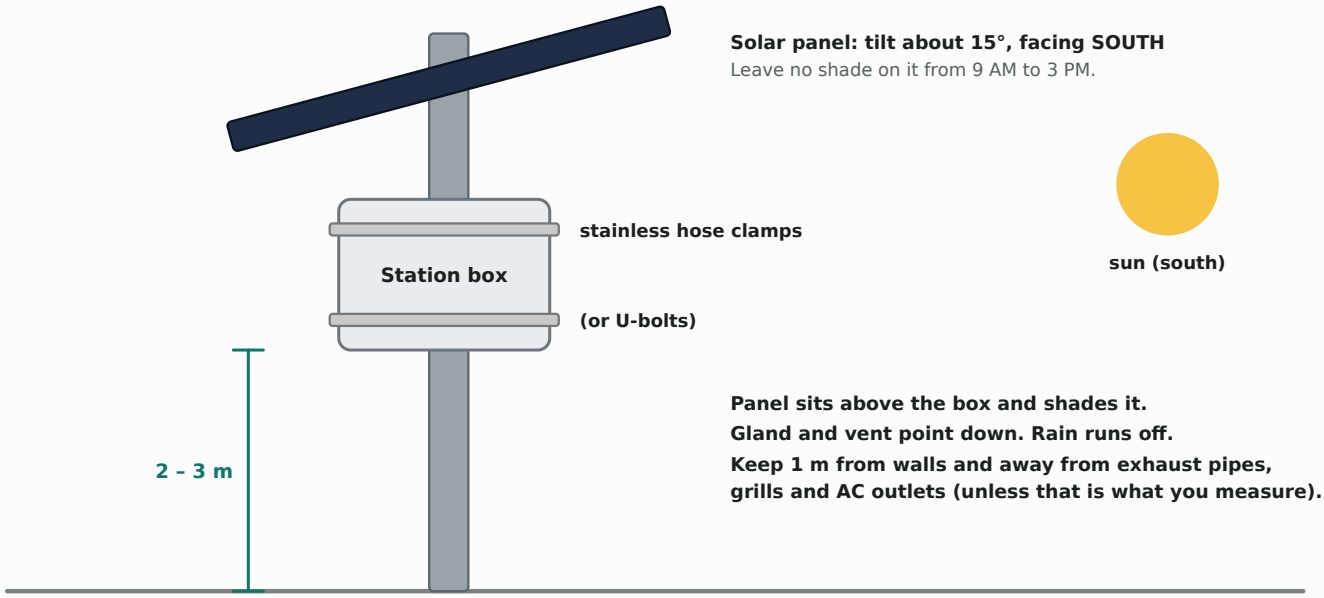


Figure 31. Mounting on a pole. The sensor should breathe the same air people do: 2 to 3 m above the ground, in open air.

1. Choose the spot with the team adviser and the barangay: open air, 2 – 3 m high, at least 1 m from walls, away from exhaust pipes, grills and air-con outlets, with Wi-Fi or mobile signal.

2. Fix the box to the pole with two stainless hose clamps (or U-bolts) through its mounting ears or plate.

3. Fix the panel above the box on a small bracket, tilted about 15°, facing south. It shades the box from the noon sun.

4 Check that the gland and vent point down. Run the solar cable along the pole with zip ties, with a drip loop before it enters the gland.

5 Write down the place, date and GPS coordinates; take photos for the paper.

STAGE 28 15 min

See it in the app and hand over

The station adds and names itself. There is nothing to type in the app.

1. Open the Ginhawa app. The station card shows new readings every minute.
2. Within 15 minutes outdoors, the GPS fixes the position and the station renames itself after its barangay. An admin can rename it in Settings → Admin.
3. Over the first day VOC and NOx readings settle (the sensor learns its surroundings for about an hour, best after a day).
4. Write down the station name and chip ID shown in the Serial Monitor for your records.

STAGE 29 ongoing

Keep it running

A short routine keeps readings accurate for months.

HOW OFTEN	WHAT	HOW
Every day (from the app)	Station online, battery level normal	Look at the card. Offline for more than 30 min: check Wi-Fi/signal and battery.
Weekly	Clean the solar panel	Wipe dust and bird droppings with a damp cloth.
Monthly	Look inside	Fuse out, open, check silica gel (replace if colour changed), loose wires, insects, water.
Monthly	Mesh and openings	Brush dust and webs off the mesh window. Never spray water into it.
After a storm	Water and mounting	Check for water inside, loose clamps, panel angle.
Every 6 – 12 months	Battery health	If the battery no longer lasts the night on sunny days, replace all 3 cells together.

BEFORE OPENING THE BOX

Always pull the fuse first. Never open the box in rain.

REFERENCE

Troubleshooting

Find the symptom and try the fixes in order.

SYMPTOM	LIKELY CAUSE	FIX
Nothing turns on	Fuse out or blown; BMS asleep; battery empty	Check the fuse with continuity. Put the panel in the sun to wake the BMS. Measure the VBAT bus.
Fuse blows at once	Short circuit or reversed wire	Cells out. VBAT to GND bus must not beep. Look for solder bridges and backwards capacitors.
I²C scan finds nothing	SDA/SCL swapped, no 3.3 V, no common GND	Swap the D21/D22 wires. Measure the 3V3 rail. Check the tap F ground link.
Only 0x40 found	SEN55 has no 5 V or wrong pin order	Measure 5 V between SEN55 pins 1 and 2. Recheck pin 1 with continuity.
Only 0x69 found	INA219 VCC or I²C wires	INA219 VCC must be 3.3 V; check its SDA/SCL wires.
Modem silent	TX/RX swapped, no VBAT, PWRKEY	Swap TXD/RXD. Measure VBAT at the modem. Press its power button if present.
Modem resets when connecting	Voltage dips in 4G bursts	Shorter, thicker VBAT/GND wires; third 1000 µF capacitor; charge the battery.
AT+CSQ = 99,99	No signal or antenna on the wrong socket	4G antenna on MAIN; try outside; try another network SIM.
No GPS fix	Antenna cannot see the sky	Test outdoors 5 minutes; GPS antenna flat side up, no metal above it.
"Connecting to Wi-Fi... failed"	Wrong name/password or 5 GHz network	Check config.h spelling; use a 2.4 GHz network.
"Enrollment failed (HTTP 401/403)"	ENROLL_SECRET differs, or station blocked	Copy it exactly from .env; ask an admin to unblock the device.
"Upload failed (HTTP -1)"	Server not reachable	Laptop on, Docker running, same Wi-Fi, correct IP in API_BASE_URL.
Readings stop at night	Battery runs out	Check charging in sun (CN3791 LED); clean the panel; remove shade.
Temperature reads too high	Box heats in the sun	Panel must shade the box; keep the SEN55 away from warm parts.
Water inside	Gland loose, holes on top, lid uneven	Dry it, re-tighten the gland, reseat the lid, fresh silica gel.

REFERENCE

Complete wiring list

Every connection in the station. With fuse and cells out, check each with continuity before the first full power-up.

FROM	TO	WIRE	BUILT IN
Holder red lead	BMS B+	18 AWG	Stage 11
Holder black lead	BMS B−	18 AWG	Stage 11
BMS P+ (or B+)	Fuse holder → INA219 VIN+	18 AWG	Stage 13
INA219 VIN−	VBAT bus tap 1	18 AWG	Stage 13
BMS P−	GND bus tap A	18 AWG	Stage 13
Solar panel + / −	CN3791 SOLAR + / −	panel cable	Stage 14
CN3791 BAT + / −	VBAT tap 2 / GND tap B	22 AWG	Stage 14
TPS63020 VIN / GND	VBAT tap 3 / GND tap C	22 AWG	Stage 15
MT3608 VIN+ / VIN−	VBAT tap 4 / GND tap D	22 AWG	Stage 16
TPS63020 VOUT / GND	ESP32 board 3V3 rail / GND rail	22 AWG	Stage 17
ESP32 3V3 pin	3V3 rail	under board	Stage 17
ESP32 GND pin	GND rail	under board	Stage 17
R3 4.7 kΩ	D21 ↔ 3V3 rail	resistor	Stage 17
R4 4.7 kΩ	D22 ↔ 3V3 rail	resistor	Stage 17
ESP32 board GND rail	Power board GND tap F	18 AWG	Stage 17
MT3608 VOUT+	SEN55 pin 1 (VDD)	22 AWG	Stage 20
MT3608 VOUT−	GND rail	22 AWG	Stage 20

FROM	TO	WIRE	BUILT IN
SEN55 pin 2 (GND), pin 5 (SEL)	GND rail	GH cable	Stage 20
SEN55 pin 3 (SDA)	ESP32 D21	GH cable	Stage 20
SEN55 pin 4 (SCL)	ESP32 D22	GH cable	Stage 20
SEN55 pin 6	not connected	—	Stage 20
INA219 VCC / GND	3V3 rail / GND rail	22 AWG	Stage 20
INA219 SDA / SCL	ESP32 D21 / D22	22 AWG	Stage 20
A7670SA VBAT / GND	VBAT tap 5 / GND tap E (+ 2 × 1000 µF)	18 AWG	Stage 21
A7670SA TXD	ESP32 RX2 (GPIO16)	22 AWG	Stage 21
A7670SA RXD	ESP32 TX2 (GPIO17)	22 AWG	Stage 21
A7670SA PWRKEY	ESP32 D4 (GPIO4)	22 AWG	Stage 21
A7670SA GND (signal)	ESP32 board GND rail	22 AWG	Stage 21
A7670SA VCCIO (if present)	3V3 rail	22 AWG	Stage 21
ESP32 VIN	nothing	—	—

REFERENCE

Parts and tools

What the team bought, and what is still needed before assembly.

Bought

- [ ] SIMCom A7670SA LTE + GNSS breakout, with 4G and GPS antennas × 1
- [ ] ESP32 DevKit V1 (already owned) × 1
- [ ] Sensirion SEN55 (already owned) × 1
- [ ] Solar panel 6 V 6 W, 270 × 180 mm × 1
- [ ] CN3791 solar charger, 6 V version × 1
- [ ] Beston 18650 65N-26, 2600 mAh (3 used + 1 spare) × 4
- [ ] BMS 1S 3.7 V 10 A × 1
- [ ] 3-cell 18650 parallel holder × 1
- [ ] TPS63020 buck-boost module, 3.3 V × 1
- [ ] MT3608 boost module × 1
- [ ] INA219 module × 1
- [ ] Blade fuses (standard size), 5 A × set
- [ ] 16 AWG waterproof inline fuse holder, standard × 1
- [ ] 18 AWG silicone wire, red and black × 2 m each
- [ ] 22 AWG silicone wire, red and black × 1 m each
- [ ] JST GH 1.25 mm 6-pin double-ended cable × 5
- [ ] 4.7 kΩ resistors × 50
- [ ] 1000 µF 25 V capacitors × 20
- [ ] PG7 cable glands, black × 5
- [ ] M12 × 1.5 breather vent, black × 5
- [ ] Perfboard 7 × 9 cm (2) and 5 × 7 cm (1) × 3
- [ ] Female pin headers 1 × 40 × 10

Still needed

- [ ] IP65 ABS junction box, about 150 × 100 × 70 mm to buy
- [ ] Nano-SIM (DITO, Smart or Globe) with data load to buy
- [ ] Stainless hose clamps or U-bolts, small panel bracket to buy
- [ ] 2 mm Allen key, 3 and 12 mm drill bits (or step bit) to buy or borrow
- [ ] Double-sided foam tape, zip ties, electrical tape, masking tape to buy
- [ ] Soldering iron, solder, flux, wick or pump buy or borrow
- [ ] Side cutter, wire stripper, pliers, small screwdrivers buy or borrow
- [ ] Multimeter with continuity beep required
- [ ] Heat gun or lighter buy or borrow
- [ ] Kitchen scale, safety glasses, hot glue, file borrow

- [ ] M3 × 10 + 6 nylon standoffs × 50
- [ ] M3 nuts × 10+
- [ ] M3 × 8 button screws × 50
- [ ] Heat-shrink tubing kit × 164 pcs
- [ ] Silica gel packs × 1 pack
- [ ] Stainless insect mesh × 1 piece

REFERENCE

Glossary

Every technical word used in this manual, in plain language.

AC / DC
Direct current (DC) flows one way, like from a battery. Everything in this station is DC. Wall sockets are AC: never connect them to the station.

AT command
A text command for the modem, always starting with AT. Example: AT+CSQ asks for the signal strength.

AWG
Wire thickness. Smaller number = thicker wire. 18 AWG carries battery power; 22 AWG is for signals and small currents.

Baud
Speed of a serial connection. The Serial Monitor must be set to the same speed as the program: 115200.

BMS
Battery Management System: the protection board on the battery.

Bus
One wire that many others connect to, so they all share the same voltage (VBAT bus, GND bus).

Capacitor
Stores a small amount of energy and releases it quickly. Electrolytic capacitors have + and − legs.

Continuity
Whether two points are electrically connected. The multimeter beeps if they are.

COM port
The name Windows gives the USB connection to the ESP32 (COM3, COM4 …).

Current (A, mA)
How much electricity flows, in amperes. 1 A = 1000 mA.

Enrollment
The station introducing itself to the app the first time it connects. It happens automatically.

Firmware
The program that runs on the ESP32.

Flux
A paste or liquid that cleans metal while soldering so the solder flows and sticks.

Fuse
A deliberate weak point that melts to cut power if too much current flows.

GND (ground)
The common return path, the "−" side of everything. All GND points in the station are connected together.

GNSS / GPS
Satellite positioning. The A7670SA finds the station's location with it.

Header
A row of pins (male) or sockets (female) at 2.54 mm spacing.

I²C
A way for chips to talk over two wires, SDA (data) and SCL (clock). The SEN55 and INA219 share these two wires. Each device has an address (0x69, 0x40).

LTE / 4G
Mobile data network. The station uses it when Wi-Fi is not available.

mAh / Wh
Battery capacity. 2600 mAh at 3.7 V is about 9.6 Wh per cell.

Module / breakout
A small ready-made board carrying a chip plus everything it needs, with labelled pins.

Pad
A flat copper area on a board where you solder.

Parallel / series
Parallel: + to +, − to −; voltage stays the same. Series: + to −; voltages add.

Polarity
Which side is + and which is −. Parts with polarity only work, or survive, one way round.

Pull-up resistor
A resistor from a signal wire to 3.3 V so the signal sits at a clean "high" when nothing is talking.

Short circuit
+ connected straight to − with nothing in between. Causes huge current, heat and fire.

Solder
A soft metal alloy melted to join wires and parts electrically.

Tinning
Coating a wire end or tip with a thin layer of solder before joining.

UART / serial
A way to send text one bit at a time over two wires, TX (transmit) and RX (receive). TX of one side goes to RX of the other.

VBAT
The battery voltage rail in this station, 3.0 – 4.2 V.

Voltage (V)
The electrical "pressure" between two points, in volts. Measured with the multimeter on V⎓.

Ginhawa capstone project. Pin assignments match firmware/ginhawa_node (SDA D21, SCL D22, modem RX2/TX2 GPIO16/17, PWRKEY GPIO4) and docs/diagrams/03-circuit-schematic.